Six axioms and their minimal geometry

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Abstract

A unified theory must explain what is observed. We begin from six characteristics of the universe that need no instrument to recognise: it closes on itself; it grows and decays; the order of its events matters; it never stands still; what is seen of it is less than what is there; and it is one thing. Taken as axioms on a compact Riemannian geometry $(T \times \Sigma) \times F$, each behavioural word given its meaning in a numbered definition, they force a unique geometry. The sixth axiom carries minimality. The time factor is the Sol torus bundle whose step is the square of the Fibonacci matrix, and a three-step time circle; the internal fibre is the pillowcase orbifold, its four cone points carrying the subalgebras $\mathbb{R}, \mathbb{C}_i, \mathbb{C}_j, \mathbb{C}_k$ of the quaternions; the dimension is twelve; the base is $\mathbb{CP}^2 \times S^2$, with Spin^c determinant line $\mathcal{O}(3)$. The fifth axiom fixes the twist of the fibre: three units of flux and one odd corner. The constants π, e, ϕ and the primes two, three and five are outputs. A minimal gauge manifold of Witten is recovered as a corollary. The principal consequence is that the twelve-dimensional manifold of the programme is derived rather than postulated.

1 What is observed, and the six axioms

There has been a long quest for a grand unified field theory. Whatever form such a theory takes, it must explain what we observe in the universe; that is the starting point, not a test applied at the end. We therefore begin with the observable characteristics of the universe in their simplest form, the ones that need no instrument and no training to recognise, and from them we state six axioms. From the six axioms we derive in this paper a twelve-dimensional manifold; later papers of the series derive the field equation on it and its consequences. The purpose of this paper is the first step: to show that the geometry on which the programme rests is forced by what is observed, rather than postulated.

1.1 What a child can observe

Six things are apparent to anyone.

I am aware of myself and of my surroundings, and I act accordingly. So does a tree, turning to the light; so does everything that lives. Whatever the universe is, it contains things that are aware of themselves, and so the universe as a whole is in some sense aware of itself: it closes on itself.

Everything changes, including the things we do not call alive: the weather, the rocks, the mountains, the continents, the earth itself. Nothing stays still.

There are cycles. Winter is followed by spring, summer, autumn and winter again; day follows night follows day. Yet no season is exactly like the last and no day repeats: the cycles return, but every cycle is different. Things grow and then fade, the seed to the tree to the fallen leaf, and the fading is as much a part of the cycle as the growing.

The order in which things are done matters, and a different order gives a different result. Anyone who has followed a recipe knows this: mix the ingredients and then cook them, and you have a cake; cook them and then mix, and you have something else. What has been done cannot be undone by doing it again in another order.

A thing and its shadow are not the same. A shadow on the ground tells a good deal about what casts it, but not everything: the shadow is flat and the thing is not. Two quite different things can cast the same shadow, and one thing can cast quite different shadows. What is seen of the world is of this kind. There is more to the world than the part of it that is seen, and what is seen does not by itself say what it is the shadow of.

A baby sees everything as one thing, and that one thing is everything. There are no labels, no categorisations, no judgements. They don't see their parents or their environment as separate from them. They exist, and all that exists is them.

1.2 The six axioms

We take these six observations as axioms, in the simplest words that carry them.

Axiom 1 (closure)**.** The universe closes on itself, so that it can be aware of itself.

Axiom 2 (growth and decay)**.** The universe grows and decays as a function of itself.

Axiom 3 (order)**.** The order of events matters, so that each instance is unique and cannot be foretold.

Axiom 4 (drive)**.** The universe is in constant change; it never stands still.

Axiom 5 (shadow)**.** We can see the shadow, but the shadow is missing information about what it is a shadow of. The same shadow can be describing two different objects, and a single object can have multiple shadows.

Axiom 6 (oneness)**.** There is only one thing, and that one thing is everything.

Closure carries the cycles and the self-awareness; growth and decay carries the seed and the leaf; order carries the recipe; drive carries the weather and the mountains; shadow carries the thing and what is seen of it; oneness carries the world of the baby. That no cycle repeats exactly is not a further axiom but a consequence of the first and the fourth together, as Section 4 shows. The fifth axiom differs in kind from the other four. They say how the universe behaves; it says that the universe has a part that is seen and a part that is not, and it is in those terms that the other four are made precise. It says also how the hidden part shows: one thing in several shadows, and several things in one. The sixth differs again. It says nothing of how the universe behaves or of how it is seen; it says that there is nothing besides it, and it is in this axiom that minimality is placed: the geometry is the least that is required to make sense of what is observed (Section 3).

1.3 Four numbers

Four numbers will appear in the derivation, and it is worth saying at the outset where each comes from, because none of them is put in by hand.

The number π is the ratio of a circle to its diameter. It enters through closure: a thing that closes on itself is a loop, and the simplest loop is a circle; and it enters a second time at the four special points of the internal geometry, which turn out to be cones whose angle is exactly half a full turn, π (Section 5).

The number $e \approx 2.718$ is the base of natural growth: a quantity that grows at a rate proportional to its own size grows as $e^{\kappa t}$. It enters through growth and decay, and nowhere else (Section 2.3).

The golden ratio $\phi = (1+\sqrt{5})/2 \approx 1.618$ is the number with $\phi^2 = \phi + 1$. It enters through the drive, as the rate at which the gentlest possible everlasting change stretches what it acts on (Section 4).

The Fibonacci series $1, 1, 2, 3, 5, 8, 13, \ldots$, in which each term is the sum of the two before it, grows at the rate ϕ: the ratio of consecutive terms tends to ϕ. It enters with the drive in a precise way. The drive, once the axioms have selected it, is the matrix $\left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, and this matrix is the square of the Fibonacci matrix $\left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, the matrix that sends a pair of consecutive Fibonacci numbers to the next pair. The least drive that never stands still is two steps of the Fibonacci recursion (Theorem 4.3).

1.4 What this paper proves, and how it is organised

The result is that the six axioms force a unique geometry:

$$\mathcal{M}_{12} \;=\; (S^1_{\mathrm{Sol}} \times S^3) \times K_8, \qquad K_8 \;=\; \left[(\mathbb{CP}^2 \times S^2) \times_w (T^2/\mathbb{Z}_2)\right]^{\mathrm{Spin}^c}. \tag{1}$$

In words: time is a circle along which a two-dimensional internal fibre is stretched and squeezed by the Fibonacci drive; the fibre, folded in half by the growth-and-decay exchange, is a pillowcase with four cone points carrying the four pieces of the quaternion algebra of the order axiom; the whole has twelve dimensions, every smaller number failing the order axiom; and the base over which the fibre sits is the sphere of the three imaginary quaternion units together with the space of their complex directions. Every discrete datum of (1), and the three primes two, three and five that appear in its arithmetic, are outputs. The symbol $\times_w$ records that the product is warped, as the second axiom requires. Which of the two pieces is scaled, and by a function on which, is not fixed in this paper: nothing derived here depends on it, and it is fixed where the warp is given its law, with the field equation.

Section 2 turns the first five sentences into mathematics, at a level intended to be followed by a first-year mathematics graduate, and isolates as numbered definitions the six points at which a behavioural word has to be given a mathematical meaning. Section 3 does the same for the sixth, in three clauses that are three further numbered choices, and defines minimality. Sections 4 to 7 derive the time factor, the fibre, the dimension and the base; Section 8 assembles the theorem; Section 9 says what the geometry dictates for the physics. From Section 4 onward the register is that of a formal mathematics paper: every load-bearing step is proved or reduced to a cited

classical theorem, the smallest case is worked by hand, and the finite and arithmetic parts are certified in Lean (Appendix A). Rigour is tagged [R] (proved, or a cited classical theorem), [D] (derived under a stated reading) and [M] (motivated).

2 The first five axioms as mathematics

This section is written to be followed without specialist background. Each axiom is a sentence about behaviour; to use it we must say what the sentence means for a shape. We do this one axiom at a time, the fifth first because the other four are stated in its terms, in words first and symbols second, and whenever a word has to be given a meaning that is a choice rather than a consequence, we say so and number the choice as a definition, so that a reader can see exactly where the axioms end and our decisions begin.

2.1 Shadow, and the shapes we consider

The axiom has two sentences: *we can see the shadow, but the shadow is missing information about what it is a shadow of*; and *the same shadow can be describing two different objects, and a single object can have multiple shadows.*

For a shape the first sentence says two things. There is more to the shape than the part along which we see: besides time and space it has a part that is not directly seen. And what is seen is what is left of the whole when that part is forgotten, the way the shadow of a solid thing on the ground is what is left of it when its height is forgotten. This is clause (i) of the definition below.

The second sentence names the two ways in which a shadow fails to tell what casts it, and each becomes a clause. That a single object can have multiple shadows is clause (ii). That the same shadow can be describing two different objects is clause (iii).

A *geometry* here is a smooth shape, of some number of dimensions, on which lengths and angles can be measured, and which is allowed to have a few isolated sharp points (cone points, like the tip of a paper cone).

Definition 2.1 (shadow, formal)**.** A geometry satisfies A5 if (i) it is a product of three pieces,

$$M \;=\; (T \times \Sigma) \times F, \tag{2}$$

where T is a one-dimensional piece carrying time, Σ is a piece carrying space, and F is an internal piece, and what is seen is the image of M under the projection

$$\pi \colon M \to T \times \Sigma, \qquad \pi(t, x, f) = (t, x),$$

that forgets F, the image under π of a point or of a subset of M being its *shadow*; and

(ii) the structure carried by the internal piece is not carried to itself by one step of the drive of the fourth axiom, so that what is seen of one thing is not the same at every step; and

(iii) the internal piece has at least two independent states at rest with the same shadow: states that differ only in the internal piece.

Clauses (ii) and (iii) are the two halves of the axiom's second sentence, taken in turn. The words *step* and *state at rest* are made exact where they are used, in Sections 4 and 7 (Definition 7.4), and it is there that the two clauses do their work (Proposition 7.6).

A product of pieces means, as for a rectangle made of two intervals, that a point of M is a point of each piece at once. Writing M this way is only a way of naming the pieces. The way lengths are measured may tie the pieces together, and the second axiom will insist that it does; and the internal piece may be carried round time by a map of itself, as Section 4 finds. That the part along which we see is time and space, and that seeing is the projection π, is the first of the choices this section records. That an object, in clauses (ii) and (iii), is a state of the internal piece at rest, and its shadow what is left of it when the internal piece is forgotten, is the second.

Remark 2.2 (what the shadow loses). The projection π is onto, and it is one-to-one only if F is a single point: if $f \neq f'$ are two points of F, then (t, x, f) and (t, x, f') are different points of M with the same shadow (t, x). The internal piece of a geometry satisfying the other four axioms is more than a point: Theorem 6.1 shows that it has dimension at least eight. In every such geometry, therefore, a point of M is not determined by its shadow. This is the second half of the sentence, and it is a consequence and not a further assumption. The axiom does not say that the whole can be recovered from the shadow. [D]

The fifth axiom is what the other four stand on. Closure will be asked of each of the three pieces; the growth law will be a tie between pieces; the order axiom will speak of operations on F; and the drive will be a flow on T together with F. None of these can be stated for a shape that has not first been divided into pieces, and in this paper it is the fifth axiom that makes the division, into what is seen and what is not. It also constrains the shape: its second and third clauses fix the twist of the internal fibre (Section 7).

2.2 Closure

The sentence is: *the universe closes on itself, so that it can be aware of itself.*

For a shape, "closes on itself" means two things. First, no piece goes on for ever and no piece has an edge: each is *compact*, like a circle or a sphere rather than a line or a disc. Time, in particular, is a circle: there is no first moment and no last. Second, there is some loop in the shape that cannot be shrunk to a point, the way the loop round a doughnut's hole cannot be shrunk but a loop drawn on a ball can. A path that returns to its start and cannot be undone is the geometric form of "aware of itself": the shape refers back to itself in a way that is part of its structure, not an accident of how it is measured.

Definition 2.3 (closure, formal)**.** A geometry satisfies A1 if T, Σ and F are compact, connected and without boundary, and the fundamental group of M is non-trivial.

The fundamental group is the collection of loops in M up to continuous deformation; "non-trivial" means some loop does not deform to a point. We ask this of the whole shape, not of each piece: the time circle already supplies such a loop, and the space and internal pieces are free to have none of their own. On any compact shape there is always a closed geodesic, a closed path that is as straight as the shape allows [11]; we ask for more, a loop that is topologically stuck, and that is the third of the choices.

2.3 Growth and decay

The sentence is: *the universe grows and decays as a function of itself.*

"As a function of itself" is the law of natural growth: a quantity w whose rate of change is proportional to its own size, $\mathrm{d}w/\mathrm{d}s = \kappa w$, and so $w = w_0 e^{\kappa s}$. This is the only place the number e enters. "Grows and decays" says that both signs of κ occur, and are two faces of one thing: there is a symmetry of the shape that swaps κ for $-\kappa$. "The universe" grows: the growth is a property of the shape, not of something placed on it, which we take to mean that the growth law sits in the way lengths are measured, as a *warp*: the size of one piece depends on where one is in another, the way the circles of latitude on a globe shrink towards the poles.

Definition 2.4 (growth and decay, formal)**.** A geometry satisfies A2 if: (i) at least one product in (2) is a warped product: the lengths measured on one piece depend on where one is in another; (ii) along the warping direction s the lengths in one direction of the warped piece are multiplied by $w = e^{\kappa s}$ and those in another by $e^{-\kappa s}$, for a constant $\kappa \neq 0$, so that $\mathrm{d}w/\mathrm{d}s = \kappa\, w$ and both signs of κ occur; (iii) there is a symmetry of the shape that swaps κ for $-\kappa$, and applied twice it is an involution ι of the warped piece.

An involution is a symmetry that undoes itself when applied twice, like a reflection. Clause (i) is the fourth of the choices: the growth is in the geometry rather than in a field placed on it. The involution ι is not identified here; Section 5 will find that it is the fold that halves the internal piece and, on the algebra of the next axiom, the operation of conjugation.

2.4 Order

The sentence is: *the order of events matters, so that each instance is unique and cannot be foretold.*

Mathematics has an exact word for "order matters": *non-commutative.* Two operations commute if doing them in either order gives the same result, as adding 2 and adding 3 do; they fail to commute if the order changes the outcome, as putting on socks and putting on shoes do. The axiom says that the operations the universe performs on its internal piece fail to commute. To make this precise we must say what those operations are.

Definition 2.5 (the fibre algebra)**.** The *fibre algebra* $\mathcal{A}$ of a geometry is the collection of operations on the fibres of F, with the product of two operations being the result of doing one and then the other, in the order in which they occur in time.

Definition 2.6 (order, formal)**.** A geometry satisfies A3 if its fibre algebra is non-commutative: there are operations a and b with $ab \neq ba$.

Two things follow at once and are not extra assumptions. Because F sits over a compact piece, there are only finitely many independent operations: $\mathcal{A}$ is finite-dimensional. Because "and then" is how operations are combined, doing a then b then c is the same whether one groups it as $(ab)c$ or $a(bc)$: $\mathcal{A}$ is associative. What the axiom does *not* say is that every operation can be undone; that it can will be a consequence of closure (Section 5), and it is that consequence,

together with a classical theorem of Frobenius [3], that will force $\mathcal{A}$ to be the quaternions, the four-dimensional number system with units $1, i, j, k$ and $i^2 = j^2 = k^2 = ijk = -1$, in which $ij = k$ but $ji = -k$. The second half of the sentence is a consequence of the first: for two observables that fail to commute, the product of the uncertainties in measuring them is bounded below by half the size of the mean of their commutator [13], so "cannot be foretold", in the quantum sense, is a theorem of "order matters". Unpredictability in the other sense, of a system whose future is sensitive to its present, belongs to the fourth axiom.

Two further definitions say *where* the fibre algebra acts, and these are the fifth and sixth choices. A shape has at every point its tangent directions, and, if it is the right kind of shape, also its spinors, the objects that carry the half-integer spin of a fermion. The order axiom will be read as saying that the non-commutative algebra is the algebra of the shape's own tangent directions and spinors, not an abstract bookkeeping device.

Definition 2.7 (the algebra acts on spinors and on tangent vectors)**.** The fibre algebra acts on the geometry at two levels: at the spinor level, on a spinor bundle over M; and at the tangent level, as an almost-quaternionic structure on M, a rank-three subbundle of $\mathrm{End}(TM)$ locally spanned by anticommuting almost-complex structures I, J, K with $IJ = K$ (the local form of the coefficient bundle of [14]), whose restriction to F carries the full algebra.

An almost-complex structure I is a rule that rotates every tangent direction by a quarter-turn, $I^2 = -1$, the way multiplication by i rotates the complex plane; a quaternionic structure is three such rules I, J, K obeying the quaternion relations. The reader need only hold on to the picture: the shape carries, at every point, a copy of the quaternion multiplication table acting on its directions.

The spinor bundle of Definition 2.7 is the complex spinor bundle, and data of the fibre algebra carried by spinors are complex-valued: the three imaginary units take values in $\mathbb{C} \otimes \mathrm{Im}\,\mathbb{H} \cong \mathbb{C}^3$. This is the tenth choice of this paper, and the seventh of this section.

Definition 2.8 (orbits are realised)**.** A homogeneous space G/H formed from the symmetry group G of the fibre algebra and the stabiliser H of a structure selected on it is *realised* by the geometry: it appears as a factor of the base over which F sits. A symmetry here is a map of the imaginary units that keeps their length and is not asked to keep their products: for the real units $x_1 i + x_2 j + x_3 k$, of length $(x_1^2 + x_2^2 + x_3^2)^{1/2}$, a real-linear map; for the complex-valued units $z_1 i + z_2 j + z_3 k$, of length $(|z_1|^2 + |z_2|^2 + |z_3|^2)^{1/2}$, a complex-linear map.

The picture here is simpler than the words. The quaternions have a symmetry group, the rotations of the three imaginary units i, j, k among themselves. If something in the geometry picks out one unit, say i, then the set of all units it *could* have picked is a sphere, and the definition says that this sphere is part of the shape, not merely a list of alternatives. That is the sixth choice, and it is what produces the two-sphere and the complex projective plane of Section 7.

2.5 Drive

The sentence is: *the universe is in constant change; it never stands still.*

For a shape with a time circle this means a flow: a rule moving every point along as time advances. "Never stands still" means the flow has no fixed points. "Constant change" means more than motion: a planet in a perfectly regular orbit moves but changes nothing, and we take the axiom to exclude that. The mathematical measure of genuine change is *entropy* of the flow: it is positive when two points that start close are driven apart, so that the far future cannot be read off from the present.

Definition 2.9 (drive, formal)**.** A geometry satisfies A4 if $T \times F$ carries a smooth flow φ_t with no fixed points and positive topological entropy.

The flow is placed on time together with the internal piece, not on time alone, because a one-dimensional circle cannot carry a flow that drives points apart; the drive is what time does to the internal piece as it goes round. This is the first point at which the order axiom and the drive act on the same object. The strong form of the drive that the construction uses, a flow that stretches in one direction and squeezes in another, is not assumed; Section 4 derives it, and shows that the gentlest such flow is the Fibonacci drive of Section 1.3.

2.6 What has been chosen

The five sentences have been turned into five conditions, and in doing so choices were made and numbered: that what is seen is the image of the whole under the projection onto time and space (Definition 2.1(i)); that an object is a state of the internal piece at rest, and its shadow what is left of it when the internal piece is forgotten (Definition 2.1(ii), (iii)); that self-reference means a loop that cannot be shrunk (Definition 2.3); that growth is in the geometry (Definition 2.4(i)); that the non-commutative algebra is the algebra of the shape's own spinors and tangent directions (Definition 2.7); and that the sphere of alternatives is part of the shape (Definition 2.8); and that the spinor bundle of Definition 2.7 is the complex spinor bundle, and data of the fibre algebra carried by spinors are complex-valued. Every result below that depends on one of these says so. Nothing in the five conditions mentions a gauge group, a particle, a generation or a coupling constant, and nothing names a factor of the manifold that will be derived. The overall size of the shape is not among the things to be fixed: under the sixth axiom, taken up in the next section, there is nothing outside the shape against which its size could be measured, and only proportions within it have a meaning.

3 The sixth axiom as mathematics: minimality

The sentence is: *there is only one thing, and that one thing is everything.*

"Everything" is read as everything that can be observed or spoken of. The axiom is not read as saying that the one thing contains all that could exist, and it does not say what, if anything, lies beyond what can be spoken of.

For a shape the sentence is given three meanings. Each is a choice of meaning and not a consequence of the words, and they are the seventh, eighth and ninth choices of this paper.

The first: nothing is left to be chosen. If the shape had a dial that could sit at different settings, something would have to set it, and there is nothing else to set it. If the axioms allowed

several versions of one piece of the shape that no relabelling carries into one another, something would have to pick one, and there is nothing else to pick.

The second: nothing is in the shape that is not needed. It carries no object, a twisting line, a larger structure, a loop, that the other five axioms do not call for.

The third: what is needed is least. The shape is the least that is required to make sense of what is observed.

One thing follows without a choice. With only one thing there is no ruler outside it. The overall size of the shape is a comparison with a second thing, and there is none: it has no meaning. A metre, a second or any other unit names a comparison between two things inside the shape, and only such proportions mean anything.

Definition 3.1 (modulus)**.** A *modulus* of a geometry satisfying A1–A5 is a continuous parameter of its structure data (the lengths, with the warp of Definition 2.4 and its rate κ, the complex and Spin^c structures, the orbifold data, the flux integers) that the six axioms and the readings of this paper leave free, taken modulo diffeomorphism and overall scale.

Definition 3.2 (oneness, formal; minimal)**.** A geometry satisfying A1–A5 satisfies A6, and is called *minimal*, if it is selected by the following three clauses. Each is applied wherever the derivation reaches a datum of which it speaks. Where two clauses would select differently, clause (a) prevails over clause (b), and clause (b) over clause (c).

(a) nothing is left to be chosen: M has no modulus, and the step of its drive (Section 4) is the only one of its trace, two steps being counted as the same when a relabelling of the torus that keeps its orientation carries one to the other, and only then;

(b) nothing is present that is not needed: M carries no object, whether a twisting line, a structure group larger than is needed, or a class of loops, that A1–A5 do not require;

(c) what is needed is least: one step of the drive of M has the least positive topological entropy; M has the least total dimension; the fibre of F carrying the drive has the least dimension and, at that dimension, the least number of handles; the twist of its fibre has the least flux; and its time circle has the least number of steps.

Clause (a) is the statement that a geometry with a dial would admit two universes differing only by that dial, with nothing to say which is the one. A clause of this kind can exclude and cannot pick: it is applied at the two points at which the first five axioms leave a datum unselected among equals, the setting of a dial and the class of a drive among those of one trace. The way of counting drives is part of the clause, and it has two parts: the mirror image of a step is another step, and a step run backwards is another step, unless a relabelling that keeps the orientation carries the one to the other. Neither part is derived from the first five axioms; both are declared here, and Section 4 shows that both are needed. Clause (b) removes what nothing forces. Clause (b) prevails over clause (c), so that clause (c) does not replace a structure that needs no further object by one with a smaller integer. On the drive clause (a) makes the selection, and clause (c) agrees with it (Section 4). That the topological integers of the geometry lie in $\{2^a 3^b\}$, the prime five entering only through the drive, is not a clause: Section 8 shows it to be an output. Overall size is not a modulus and is not selected: it has no meaning.

Definition 3.3 (the same geometry; unique)**.** Two geometries satisfying A1–A5 are *the same* when a relabelling carries one to the other: a one-to-one map of the one shape onto the other, smooth in both directions away from the cone points, that carries T to T, Σ to Σ, F to F and cone points to cone points; carries the direction of time of the one to that of the other, and the handedness that Definition 2.7 fixes on the internal piece of the one to that of the other; and carries every datum of the one to that of the other, the factors, the discrete data and the structure data of Definition 3.1, lengths being compared up to one positive constant, since the overall size has no meaning. A geometry of a given kind is *unique* when any two of that kind are the same.

Remark 3.4 (mirror image and run backwards)*.* The mirror image of a geometry, and a geometry run backwards, are therefore other geometries, unless a relabelling of Definition 3.3 carries the one to the other. For $\mathcal{M}_{12}$ run backwards, the quarter-turn S of the fibre carries the step, the lengths and the corner signs to themselves, since $SM_0S^{-1} = M_0^{-1}$; that it carries every datum of $\mathcal{M}_{12}$ to itself is not shown here. Whether the geometry has a symmetry that mirrors the torus is not shown here.

4 The drive

This section takes up the drive, under three headings: hyperbolicity, the least drive and its uniqueness, and worked cases.

4.1 Hyperbolicity

Let the fibre of F carrying the drive be a torus $T^2 = \mathbb{R}^2/\mathbb{Z}^2$; that it is one is proved in Section 5 (Proposition 5.5), and nothing in the present section is used there. The drive of A4 acts on T^2 by its return map, an element $M \in \mathrm{SL}_2(\mathbb{Z})$ with trace t. One application of this map is a *step* of the drive. Whether one circuit of the time circle is one step or several is not assumed in this section; Corollary 7.8 settles it.

Time is a circle, and the internal piece is carried round it by a map of itself. That map is a whole number of applications of one map, the step of the drive; how many is settled by Corollary 7.8.

The flow of Definition 2.9 moves the fibre of F that carries the drive and nothing else of the internal piece: on the rest of F it is the identity. No axiom asks the drive to move more. This is a choice and is not derived.

A step of the drive keeps the handedness that Definition 2.7 fixes on the internal piece: as a map of $T^2 = \mathbb{R}^2/\mathbb{Z}^2$ it preserves orientation, and its matrix M has determinant $+1$, so that $M \in \mathrm{SL}_2(\mathbb{Z})$. The Fibonacci matrix $F = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ has determinant -1: it reverses the handedness and is not a step. Its square $M_0 = F^2 = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ has determinant $(-1)^2 = +1$ and is the least power of F that is a step.

The step of the drive is taken to be the linear map of its matrix: for $M \in \mathrm{SL}_2(\mathbb{Z})$ it is the map $x \mapsto Mx \bmod \mathbb{Z}^2$ of $T^2 = \mathbb{R}^2/\mathbb{Z}^2$. Definition 2.9 asks only for a smooth flow. That the step is the linear one is a choice and is not derived.

Proposition 4.1 (A4 is hyperbolicity)**.** *A4 holds for the return map M if and only if $|t| > 2$.* [R]

Proof. The characteristic polynomial is $\lambda^2 - t\lambda + 1$.

If $|t| < 2$ the eigenvalues are complex of modulus one and M has finite order: every point is periodic and the entropy vanishes, contradicting Definition 2.9.

If $|t| = 2$ the eigenvalues are ± 1 and M is a shear: orbits separate linearly, the entropy vanishes.

If $|t| > 2$ the eigenvalues are real, λ and λ^{-1} with $|\lambda| > 1$; M is Anosov, its entropy is $\log|\lambda| > 0$, and the induced flow on the mapping torus has no fixed points [8].

Let $M \in \mathrm{SL}_2(\mathbb{Z})$ have trace t with $|t| > 2$.

A point of $T^2 = \mathbb{R}^2/\mathbb{Z}^2$ is carried back to itself by some number $n \geq 1$ of steps of M if and only if both its coordinates are rational.

No other point of the torus ever returns. □

4.2 The least drive and its uniqueness

Clause (a) of Definition 3.2 makes the selection: trace three is the only trace at which there is one drive and not several (Theorem 4.2, Proposition 4.5). The trace is taken positive. This is an input and not a selection: $-M$ is M followed by $z \mapsto -z$, which Section 5 finds to be the fold, so that M and $-M$ have the same entropy and are the same map of the folded fibre. Clause (c), which is applied to the step, agrees: the least entropy is that of the least $|t| > 2$, which is $t = 3$ (certificate `L1_trace3_min`). The content of the section is that there is one drive of trace three, and that at every larger trace there is more than one.

The selection is not that trace three is the only hyperbolic trace, which is false, but that it is the least, and that at the least trace there is nothing to choose: the theorem says that the gentlest drive which still drives is a single object up to relabelling the torus. The three proofs are independent in kind, a direct descent, one through ideal classes and one through reduced forms, and they are followed by the explicit relabellings that show the uniqueness at work.

Theorem 4.2 (the drive)**.** *Every $M \in \mathrm{SL}_2(\mathbb{Z})$ of trace three is conjugate in $\mathrm{SL}_2(\mathbb{Z})$ to $M_0 = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$. Consequently the minimal drive is unique; its eigenvalues are $\phi^{\pm 2}$ with $\phi = (1+\sqrt{5})/2$, its entropy is $2\log\phi$, its splitting field is $\mathbb{Q}(\sqrt{5})$, and the time factor of a minimal geometry is the Sol torus bundle S^1_{Sol} with this map as its step.* [R]

Proof. Three routes.

The first is a direct descent and is complete in itself.

Let $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ have determinant one and trace three, so that $4bc = 5 - (2a-3)^2$.

The right side is congruent to four modulo eight, so b and c are odd.

Conjugation by $\left(\begin{smallmatrix} 1 & k \\ 0 & 1 \end{smallmatrix}\right)$ replaces a by $a + kc$ and leaves c unchanged; choose k so that $|2a-3| \leq |c|$.

If $|c| \geq 3$ then $(2a-3)^2 \geq 9$, since otherwise $|bc| = 1$; so $4|b||c| = (2a-3)^2 - 5 < c^2$ and $|b| < |c|$.

Conjugation by the quarter-turn $S = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ puts $-b$ in the place of c.

Repeating, we reach $|c| = 1$, where $2a - 3 = \pm 1$ and $b = c$.

A last shift of a gives M_0 when $c = 1$ and M_0^{-1} when $c = -1$, and $SM_0S^{-1} = M_0^{-1}$.

There is a single class.

Second, by the theorem of Latimer and MacDuffee [9] as sharpened by Taussky [18], the classes of integer matrices with characteristic polynomial p, under conjugation by integer matrices of determinant ± 1, are in bijection with the ideal classes of the order $\mathbb{Z}[\lambda]$, λ a root of p.

Here $p = \lambda^2 - 3\lambda + 1$ with root $\phi^2 = \phi + 1$, so $\mathbb{Z}[\phi^2] = \mathbb{Z}[\phi]$, the ring of integers of $\mathbb{Q}(\sqrt{5})$, which has class number one.

There is a single class under conjugation of determinant ± 1.

Conjugation by $\left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$ carries M_0 to $\left(\begin{smallmatrix} 2 & -1 \\ -1 & 1 \end{smallmatrix}\right)$, which the descent carries back to M_0, so there is a single class under $\mathrm{SL}_2(\mathbb{Z})$ as well.

Third, by the classical correspondence between hyperbolic conjugacy classes in $\mathrm{SL}_2(\mathbb{Z})$ and classes of indefinite binary quadratic forms [4, 15], under which $M = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ of trace t corresponds to the form $cx^2 + (d - a)xy - by^2$ of discriminant $t^2 - 4$.

At $t = 3$ the discriminant is five, squarefree, so all forms are primitive, and a form is reduced when $0 < b < \sqrt{5}$ and $\sqrt{5} - b < 2|a| < \sqrt{5} + b$.

Over the integers these inequalities force $(a, b, c) \in \{(1, 1, -1), (-1, 1, 1)\}$ (certificate `L1_reduced_disc5`), the two forms lie on one Gauss cycle (`L1_cycle_disc5`), and classes correspond to cycles.

Again a single class.

Smallest case by hand.

The candidates M_0 and $M_0^{-1} = \left(\begin{smallmatrix} 1 & -1 \\ -1 & 2 \end{smallmatrix}\right)$ are conjugate by the quarter-turn $S = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$: $SM_0S^{-1} = M_0^{-1}$ (`L1_conj_M0_inv`); M_0 is symmetric, so its transpose adds nothing.

The eigenvalue and field statements are the characteristic equation; the Möbius action $\tau \mapsto (2\tau + 1)/(\tau + 1)$ has fixed points solving $\tau^2 - \tau - 1 = 0$, namely ϕ and $-1/\phi$. □

4.3 Worked cases

Three trace-three matrices and their conjugators. The theorem asserts that any $A \in \mathrm{SL}_2(\mathbb{Z})$ of trace three is SM_0S^{-1} for some S.

Three instances, each checked by multiplying out:

$$A = \begin{pmatrix} 3 & -1 \\ 1 & 0 \end{pmatrix}, \qquad S = \begin{pmatrix} -2 & 1 \\ -1 & 0 \end{pmatrix};$$

$$A = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}, \qquad S = \begin{pmatrix} -1 & -1 \\ 2 & 1 \end{pmatrix};$$

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 3 \end{pmatrix}, \qquad S = \begin{pmatrix} -2 & -1 \\ 3 & 1 \end{pmatrix}.$$

In each case $\det S = 1$ and $SAS^{-1} = M_0$; the second instance is M_0^{-1} with a conjugator different from the quarter-turn of the proof, showing that the conjugator is not unique while the class is.

The reduction of a form. The form of $A = \left(\begin{smallmatrix} 3 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ is $cx^2 + (d-a)xy - by^2 = x^2 - 3xy + y^2$, discriminant $9 - 4 = 5$.

It is not reduced ($b = -3 < 0$).

The Gauss step $\rho(a, b, c) = (c, b', c')$ with $b' \equiv -b \pmod{2c}$ and b' in the reduced window gives $(1, -3, 1) \mapsto (1, 1, -1)$, one of the two reduced forms; the next step gives $(-1, 1, 1)$ and the next returns to $(1, 1, -1)$.

The cycle has length two and contains both reduced forms, so there is one class.

The eigenvalues and the entropy. $\lambda^2 - 3\lambda + 1 = 0$ gives $\lambda = (3 \pm \sqrt{5})/2$; since $\phi^2 = \phi + 1 = (3 + \sqrt{5})/2$, the eigenvalues are $\phi^2 \approx 2.618$ and $\phi^{-2} \approx 0.382$, and the entropy is $\log \phi^2 = 2 \log \phi \approx 0.962$.

For comparison the next admissible trace, four, has eigenvalue $2 + \sqrt{3} \approx 3.732$ and entropy ≈ 1.317: larger, as clause (c) requires it to be, and with splitting field $\mathbb{Q}(\sqrt{3})$ rather than $\mathbb{Q}(\sqrt{5})$, so that the prime five is tied to the minimal choice.

The half-lattice evaluations. With $M_0 = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$: $(0,0) \mapsto (0,0)$; $(\frac{1}{2}, 0) \mapsto (1, \frac{1}{2}) \equiv (0, \frac{1}{2})$; $(0, \frac{1}{2}) \mapsto (\frac{1}{2}, \frac{1}{2})$; $(\frac{1}{2}, \frac{1}{2}) \mapsto (\frac{3}{2}, 1) \equiv (\frac{1}{2}, 0)$.

Reduced mod two, $\bar{M}_0 = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, $\bar{M}_0^2 = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, $\bar{M}_0^3 = I$.

The even-trace matrix $\left(\begin{smallmatrix} 2 & 1 \\ 3 & 2 \end{smallmatrix}\right)$ reduces to $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, a transposition of order two: it swaps two corners and fixes two, and cannot cycle three.

Theorem 4.3 (the Fibonacci drive)**.** *Let $F = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ be the Fibonacci matrix, so that $F\binom{F_n}{F_{n-1}} = \binom{F_{n+1}}{F_n}$ and $F^n = \left(\begin{smallmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{smallmatrix}\right)$. Then $M_0 = F^2$. The minimal drive is two steps of the Fibonacci recursion; its eigenvalue ϕ^2 is the square of the Fibonacci growth rate ϕ, and F itself, having determinant -1 and trace 1, is not an element of $\mathrm{SL}_2(\mathbb{Z})$: M_0 is the least power of F that is.* [R]

Proof. $F^2 = \left(\begin{smallmatrix} 1\cdot1+1\cdot1 & 1\cdot1+1\cdot0 \\ 1\cdot1+0\cdot1 & 1\cdot1+0\cdot0 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ (certificate `L1_M0_is_Fib_squared`).

The entry recursion $(XF)_{11} = X_{11} + X_{12}$ is the Fibonacci law (`L1_Fib_recursion`), and $F^2, \ldots, F^6$ carry $(2,1,1,1)$, $(3,2,2,1)$, $(5,3,3,2)$, $(8,5,5,3)$, $(13,8,8,5)$ (`L1_Fib_powers`).

The characteristic polynomial of F is $\lambda^2 - \lambda - 1$ with roots $\phi, -1/\phi$; squaring gives ϕ^2, ϕ^{-2}, the eigenvalues of Theorem 4.2.

$\det F = -1$, $\operatorname{tr} F = 1$, $\det F^2 = 1$, $\operatorname{tr} F^2 = 3$ (`L1_Fib_not_SL2`). □

The theorem says in what sense the golden ratio and the Fibonacci series are forced rather than found: the least everlasting change that the axioms allow is the Fibonacci recursion applied twice, and ϕ is its growth rate. That the cat map is the square of the Fibonacci matrix is classical; what is new is that the axioms select it.

Proposition 4.4 (the corner three-cycle)**.** *For $M \in \mathrm{SL}_2(\mathbb{Z})$, the reduction $\bar{M} \in \mathrm{GL}_2(\mathbb{F}_2) \cong S_3$ has order three if and only if t is odd. For M_0, $\bar{M}_0 = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ has order three; on the half-lattice*

points $P_1 = (0,0)$, $P_2 = (\frac{1}{2},0)$, $P_3 = (0,\frac{1}{2})$, $P_4 = (\frac{1}{2},\frac{1}{2})$ *the map* M_0 *fixes* P_1 *and permutes the others as the cycle* $(P_2\,P_3\,P_4)$. *[*R*]*

Proof. Over $\mathbb{F}_2$ the characteristic polynomial is $\lambda^2 + \bar{t}\lambda + 1$: for even t it is $(\lambda+1)^2$ and $\bar{M}$ has order one or two; for odd t it is $\lambda^2 + \lambda + 1$, irreducible, with roots of order three in $\mathbb{F}_4^\times$.

The finite check over all of $\mathrm{SL}_2(\mathbb{F}_2)$ is `L1_odd_trace_iff_order_three`.

The action on the half-lattice: $M_0P_1 = P_1$; $M_0(\frac{1}{2},0) = (1,\frac{1}{2}) \equiv P_3$; $M_0(0,\frac{1}{2}) = (\frac{1}{2},\frac{1}{2}) = P_4$; $M_0(\frac{1}{2},\frac{1}{2}) = (\frac{3}{2},1) \equiv P_2$. □

The proposition is used in Section 5 to assign the corner algebras. We note that the orientation of the three-cycle, $P_2 \to P_3 \to P_4$ against $P_2 \to P_4 \to P_3$, is the difference between M_0 and M_0^{-1}, which Theorem 4.2 shows to be a single conjugacy class: the orientation is a convention with no invariant content (Appendix B).

Proposition 4.5 (one drive at trace three only)**.** *For every integer* $t \geq 4$ *the matrix* $A_t = \left(\begin{smallmatrix} t & -1 \\ 1 & 0 \end{smallmatrix}\right)$ *of trace* t *is not conjugate in* $\mathrm{SL}_2(\mathbb{Z})$ *to its inverse, so that there are at least two classes of trace* t. *At* $t = 3$ *there is one class (Theorem 4.2). Among positive traces, and with drives counted as in clause (a) of Definition 3.2, trace three is therefore the only one at which the drive is a single object, and the clause selects it. [*R*]*

Proof. $A_t^{-1} = \left(\begin{smallmatrix} 0 & 1 \\ -1 & t \end{smallmatrix}\right)$.

Let $P = \left(\begin{smallmatrix} p & q \\ r & s \end{smallmatrix}\right)$ be an integer matrix with $PA_t = A_t^{-1}P$.

Comparing entries gives $r = tp + q$ and $s = -p$, so that $\det P = -(p^2 + tpq + q^2)$ (certificate `L7_det_of_intertwiner`).

The two matrices are therefore conjugate in $\mathrm{SL}_2(\mathbb{Z})$ exactly when $p^2 + tpq + q^2 = -1$ has a solution in integers.

Suppose it has one.

Then $pq < 0$, since otherwise the left side is not negative.

The equation is unchanged when the signs of p and q are both reversed and when the two are exchanged, so we may write $p = x$ and $q = -y$ with $x \geq y \geq 1$ and $x^2 - txy + y^2 = -1$, and take such a pair with $x + y$ least.

If $x > y$, the second root of $X^2 - tyX + (y^2+1)$ is $x' = ty - x = (y^2+1)/x$, a positive integer, and $x' \leq (y^2+1)/(y+1) \leq y$.

Then (y, x') is a pair of the same kind with a smaller sum, which contradicts the choice.

So $x = y$, and $x^2(2-t) = -1$ gives $x = 1$ and $t = 3$ (`L7_trace3_solution`).

For $t \geq 4$ there is no solution (`L7_descent`, `L7_no_int_solution`, `L7_not_conj_inverse`).

The hypothesis on the relabellings is used: $P = \left(\begin{smallmatrix} 1 & 0 \\ t & -1 \end{smallmatrix}\right)$ has determinant -1 and satisfies $PA_t = A_t^{-1}P$ for every t (`L7_flip`), so that under relabellings that reverse the orientation the drive is conjugate to its inverse at every trace and the argument selects nothing.

The other part of the count is used as well: if a step and its inverse are counted as one, A_t and A_t^{-1} are one drive at every trace, with no relabelling at all. □

The two parts of the count of clause (a) are thus both needed, and a computation shows how much rests on them. Counting the classes of each trace up to sixty by the cycles of reduced forms [4], in four ways, gives the following. With mirror images counted as different and a step run backwards counted as different, trace three alone has a single drive. If either is counted as the same, or both, the traces 3, 4, 5, 9 and 21 each have a single drive; at the last four the two drives of the first count are mirror images of each other and are also each other run backwards. In every one of the four counts the least trace with a single drive is three, so that the drive of the theorem does not depend on the count; what depends on it is whether clause (a) or clause (c) is the ground.

The first five axioms do not supply either part. The fold, the corners and their algebras, the corner signs, the flux of the twist and the element of the drive on spinors, as this paper defines them, are each carried to themselves by a mirror of the torus and by the reversal of the drive; Definition 2.7 fixes a handedness on the internal piece, but it is one of the numbered choices, and it does not say whether the whole geometry has a symmetry that mirrors the torus. The count would fail if the geometry had such a symmetry, or if a later paper had to count a state and the same state run backwards as one.

5 The fibre

This section takes up the fibre, under six headings: its algebra, the four corners, the surface, the collapse at the corners, worked cases, and the pillowcase.

5.1 The fibre algebra is the quaternions

Lemma 5.1 (the norm form)**.** *Let $\mathcal{A}$ be an associative real algebra with unit $1 \neq 0$, and let ι act on it as an anti-involution $a \mapsto a^*$ (the algebraic image of exchanging growth and decay: composition run backwards): a real-linear map with $(ab)^* = b^*a^*$ and $(a^*)^* = a$ for all a, b. Suppose its fixed elements are exactly the real multiples of 1. Then every $a \in \mathcal{A}$ is $x + u$ with x real and $u^* = -u$; for every such u the square u^2 is real; and $N(a) = aa^* = a^*a = x^2 - u^2$ is real. For $u \neq 0$ with $u^* = -u$ the elements 1 and u are linearly independent, and the unit group of $\mathbb{R} \oplus \mathbb{R}u$ is the curve $\{x + yu : x^2 - u^2y^2 = 1\}$ in the plane with coordinates (x, y). Moreover $N(a) > 0$ for every $a \neq 0$ if and only if this curve is compact for every $u \neq 0$ with $u^* = -u$. [R]*

Proof. With $x = (a + a^*)/2$ and $u = (a - a^*)/2$, by linearity and $(a^*)^* = a$, $x^* = x$, so x is real, and $u^* = -u$, and $a^* = x - u$; $(u^2)^* = u^*u^* = (-u)(-u) = u^2$, so u^2 is fixed and therefore real; x is real and commutes with u, so $aa^* = (x+u)(x-u) = x^2-u^2$ and $a^*a = (x-u)(x+u) = x^2-u^2$; if $u = r \cdot 1$ for a real r then $u^* = u$, so $u = -u$ and $u = 0$, hence for $u \neq 0$ the elements 1 and u are independent and $(x, y) \mapsto x + yu$ is one-to-one; the curve $x^2 - u^2y^2 = 1$ is an ellipse if $u^2 < 0$, the two lines $x = \pm 1$ if $u^2 = 0$ and a hyperbola if $u^2 > 0$, so it is compact (closed

and bounded in the plane) exactly when $u^2 < 0$; if $u^2 < 0$ for every $u \neq 0$ with $u^* = -u$ then $N(a) = x^2 - u^2 > 0$ unless $x = 0$ and $u = 0$; conversely $N(u) = -u^2 > 0$ gives $u^2 < 0$. □

The proposition that follows is the place where the first axiom does algebraic work. Non-commutativity alone admits the split quaternions, in which two of the three imaginary units square to $+1$ and whose unit groups are hyperbolas.

Proposition 5.2 ($\mathcal{A} = \mathbb{H}$)**.** *Under A1, A2 and A3 the fibre algebra is the quaternions. [*D*]*

Proof. By Definition 2.5 $\mathcal{A}$ is finite-dimensional and associative.

The fibre algebra has a unit $1 \neq 0$, and ι acts on it as an anti-involution $a \mapsto a^*$ whose fixed elements are exactly the real multiples of 1. This is a reading and is not derived.

A1 requires every factor to close, and this is read as asking that the unit group of $\mathbb{R} \oplus \mathbb{R}u$ be compact for every $u \neq 0$ of $\mathcal{A}$ with $u^* = -u$.

By Lemma 5.1, $N(a) > 0$ for every $a \neq 0$, and $a \cdot a^*/N(a) = a^*/N(a) \cdot a = 1$: every non-zero element of $\mathcal{A}$ has an inverse, and $\mathcal{A}$ is a division algebra.

By Frobenius [3] a finite-dimensional associative real division algebra is $\mathbb{R}$, $\mathbb{C}$ or $\mathbb{H}$, and A3 excludes the two commutative cases.

The alternative that compactness excludes is the split algebra $M_2(\mathbb{R})$, which carries the same $\pm 1, \pm i, \pm j, \pm k$ pattern with non-compact unit groups on two of its three imaginary lines.

The relations $i^2 = j^2 = k^2 = ijk = -1$ and the non-commutativity $ij \neq ji$ are certified in an integer model (`L2_hamilton`, `L2_noncomm`), as is positive-definiteness of $N(q) = q\bar{q}$ (`L2_norm_posdef`). □

5.2 The four corners

Proposition 5.3 (corner count)**.** *The subalgebras of $\mathbb{H}$ generated by a single unit of $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ are $\mathbb{R}$, $\mathbb{C}_i$, $\mathbb{C}_j$, $\mathbb{C}_k$, in bijection with $Q_8/\{\pm 1\} \cong \mathbb{Z}_2 \times \mathbb{Z}_2$: there are exactly four. The group of inner automorphisms by units, $V = \{\mathrm{id}, \mathrm{Ad}_i, \mathrm{Ad}_j, \mathrm{Ad}_k\}$ with $\mathrm{Ad}_u(q) = uqu^{-1}$, is the same Klein four-group. [*R*]*

Proof. u and $-u$ generate the same line; four lines, four classes (`L2_corner_count`).

$\mathrm{Ad}_u = \mathrm{Ad}_{-u}$ and $\mathrm{Ad}_i \mathrm{Ad}_j = \mathrm{Ad}_k$ by Hamilton's relations. □

Proposition 5.4 (fixed subalgebras)**.** *Let ι act on $\mathbb{H}$ as conjugation $q \mapsto \bar{q}$. Among the eight elements of $V \times \langle \iota \rangle$, those whose fixed set in $\mathbb{H}$ is a commutative subalgebra are exactly ι, Ad_i, Ad_j, Ad_k, with fixed sets $\mathbb{R}$, $\mathbb{C}_i$, $\mathbb{C}_j$, $\mathbb{C}_k$ respectively;* id *fixes all of $\mathbb{H}$, which is not commutative, and $\iota \circ \mathrm{Ad}_u$ fixes the three-dimensional $u^\perp$, which is not closed under multiplication. [*R*]*

Proof. Direct computation on $q = a + bi + cj + dk$: $\bar{q} = q$ iff $b = c = d = 0$; $\mathrm{Ad}_i\, q = a + bi - cj - dk = q$ iff $c = d = 0$, and cyclically; $\overline{\mathrm{Ad}_i\, q} = q$ iff $b = 0$, and $jk = i \notin i^\perp$.

Certified in `L2_fix_conj`, `L2_fix_Ad_i`, `L2_fix_Ad_j`, `L2_fix_Ad_k`, `L2_fix_conj_Ad_i`, `L2_iperp_not_closed`. □

5.3 The fibre is a surface, and a torus

Proposition 5.5 (the fibre)**.** *Let the fibre X of F carrying the drive be a compact connected manifold without boundary, and let the fold ι be a continuous involution of X with exactly four fixed points, each isolated. Under clause (c) of Definition 3.2 the fibre is the torus T^2, on which ι is inversion $z \mapsto -z$ and the drive is a hyperbolic element of* $\mathrm{SL}_2(\mathbb{Z})$*. [D]*

Proof. Let the fibre X of F carrying the drive be a compact connected manifold without boundary, and let the fold ι be a continuous involution of X with exactly four fixed points, each isolated.

If X has dimension zero it is a point, and ι has one fixed point.

If X has dimension one it is a circle.

Suppose ι fixes a point p of it.

Then ι maps the open arc $X \setminus \{p\}$ to itself, one-to-one and continuously, and so is increasing or decreasing there.

If it is increasing it is the identity, since $\iota(x) > x$ would give $x = \iota(\iota(x)) > \iota(x)$ and $\iota(x) < x$ would give $x = \iota(\iota(x)) < \iota(x)$; its fixed points are then not isolated, which the hypothesis excludes.

If it is decreasing it fixes exactly one point of the arc, and ι has exactly two fixed points.

So ι has no fixed point or exactly two, and not four.

No such X has dimension less than two, and the torus T^2 with $z \mapsto -z$ is one of dimension two.

Under clause (c) of Definition 3.2 the fibre is a surface.

An involution of a surface with isolated fixed points is a half-turn near each of them, so it preserves orientation.

Let X be a compact connected surface without boundary that carries a handedness, and let ι be an involution of X that keeps the handedness and has exactly four fixed points, each isolated.

Then X/ι is a compact connected surface without boundary.

Take a triangulation of X/ι having the images of the four fixed points among its vertices: every cell of it is the image of two cells of X, except those four vertices, each of which is the image of one.

Hence $\chi(X) = 2\chi(X/\iota) - 4$.

Since $\chi(X/\iota) \leq 2$, with equality only when X/ι is a sphere, $\chi(X) \leq 0$: X is not a sphere, and $\chi(X) = 0$ exactly when X/ι is a sphere.

A compact connected surface without boundary that carries a handedness and has $\chi = 0$ is the torus T^2, which has one handle, and every other such surface with $\chi < 0$ has more than one.

The torus $T^2 = \mathbb{R}^2/\mathbb{Z}^2$ carries such an involution: $z \mapsto -z$ fixes exactly the four points with $2z \in \mathbb{Z}^2$.

Under clause (c) of Definition 3.2 the fibre is therefore the torus.

The drive is a map of the folded fibre T^2/ι.

A one-to-one continuous map f of T^2 that gives a map of T^2/ι carries each pair $\{x, \iota x\}$ to such a pair, so that $f(\iota x)$ is $f(x)$ or $\iota f(x)$.

Where $\iota x \neq x$ it is not $f(x)$, since f is one-to-one, and so $f(\iota x) = \iota f(x)$.

The fixed points of ι are isolated, and by continuity $f\iota = \iota f$ on the whole of T^2: the drive commutes with the fold.

For the torus the quotient Q is then an orbifold with k cone points of order two and $0 = \chi(T^2) = 2(\chi(Q) - k/2)$, so Q is a sphere with four cone points, the orbifold $(2, 2, 2, 2)$, which is unique [19]; its double cover by the torus is unique, and ι is inversion.

Inversion of T^n has 2^n fixed points (`L2_two_torsion`), so four corners agree with $n = 2$. □

That ι has exactly four isolated fixed points is Proposition 5.3 read geometrically: the fixed points of ι are the points at which the algebra reduces, and the next proposition makes the reduction precise.

5.4 The collapse at the corners

Proposition 5.6 (the corner algebras)**.** *Let ι and M_0 act on T^2 with ι inversion and M_0 the drive of Theorem 4.2, and let the algebra at a point be the subalgebra of $\mathbb{H}$ fixed by those of the fold and one step of the drive that leave the point in place, the local form of ι being conjugation at the drive-fixed corner and the inner automorphism by the surviving unit at the three cycled corners, and the drive acting as the order-three inner automorphism Ad_w, $w = \frac{1}{2}(1 + i + j + k)$. Then the algebra is $\mathbb{R}$ at P_1, $\mathbb{C}_i$, $\mathbb{C}_j$, $\mathbb{C}_k$ at the three other fixed points of ι, permuted cyclically by the drive, and $\mathbb{H}$ at every other point. [D]*

Proof. The principle is A4 read on A3: order has nothing to act on where nothing moves, and the surviving algebra at a point is the subalgebra of $\mathbb{H}$ fixed by those of the fold and one step of the drive that leave the point in place.

P_1 is fixed by ι and by M_0 (Proposition 4.4); the subalgebra fixed by conjugation is $\mathbb{R}$ (Proposition 5.4), and $\mathbb{R}$ is central, so it is fixed by Ad_w as well.

The points P_2, P_3, P_4 are fixed by ι and permuted by M_0; the local form of ι at such a point is the inner automorphism by the surviving unit, and its fixed subalgebra is the corresponding complex line.

The drive carries the lines with the points: Ad_w sends $i \mapsto j \mapsto k \mapsto i$ (certificate `L2_Ad_w_cycles`, with $(2w)^3 = -8$ giving order three, `L2_w_cubed`), so $\mathrm{Ad}_w \mathrm{Ad}_i \mathrm{Ad}_w^{-1} = \mathrm{Ad}_j$ and $\mathbb{C}_i \mapsto \mathbb{C}_j \mapsto \mathbb{C}_k$ exactly as $P_2 \to P_3 \to P_4$.

A point that neither the fold nor one step of the drive leaves in place has nothing fixed and carries $\mathbb{H}$. □

The reading in the statement, that the local form of ι is conjugation at the drive-fixed corner and the inner automorphism by the surviving unit at the three cycled corners, is the one place in

this section where the proposition is [D] rather than [R]. It is the statement that the involution's action on the algebra varies by corner in the way the drive's three-cycle requires, and it is consistent with the corner lift of Corollary 7.7, which distinguishes one corner by a sign; but the lift acts on spinors and the reading acts on the algebra, and the identification of the one with the other is a definition rather than a computation. We state it as such. The labelling of which cycled corner carries which unit is a convention and is fixed once in Appendix B.

5.5 Worked cases

The Hamilton product. For $p = p_1 i + p_2 j + p_3 k$ and $q = q_1 i + q_2 j + q_3 k$ purely imaginary, $pq = -(\mathbf{p} \cdot \mathbf{q}) + \mathbf{p} \times \mathbf{q}$: the real part is minus the dot product, the imaginary part the cross product.

The smallest case is $p = q = i$: $i^2 = -1$, the whole product real.

The imaginary part alone is not closed under multiplication, which is why the fibre algebra is all of $\mathbb{H} = \mathbb{R} \oplus \operatorname{Im}\mathbb{H}$ and not its imaginary part.

Compact and split corners. On $\mathbb{C}_u = \mathbb{R} \oplus \mathbb{R}u$ the norm is $N(x + yu) = x^2 - u^2 y^2$.

For $u = i$, $N = x^2 + y^2$ and the unit group $N = 1$ is the circle $U(1)$.

In the split algebra $M_2(\mathbb{R})$, with $j^2 = +1$, $N = x^2 - y^2$ on $\mathbb{C}_j$ and the unit group is the hyperbola $\{x^2 - y^2 = 1\}$, non-compact; the element $1 + j$ has $N = 0$ and no inverse.

A1 excludes the second picture at every corner, and with it every zero-divisor.

Fixed sets, by coordinates. For $q = a + bi + cj + dk$: $\bar{q} = a - bi - cj - dk$, equal to q iff $b = c = d = 0$.

$\mathrm{Ad}_i\, q = iqi^{-1} = -iqi = a + bi - cj - dk$, equal to q iff $c = d = 0$.

$\overline{\mathrm{Ad}_i\, q} = a - bi + cj + dk$, equal to q iff $b = 0$: the fixed set $\{a + cj + dk\}$ contains j and k but $jk = i$, so it is not a subalgebra.

Right-multiplication by i composed with conjugation, $q \mapsto \overline{qi}$, fixes only the line $\mathbb{R}(1 - i)$, which is not closed either ($(1 - i)^2 = -2i$).

The four commutative fixed subalgebras are therefore exactly those of Proposition 5.4, and nothing else of the eight candidates qualifies.

The order-three automorphism. With $2w = 1 + i + j + k$ and $\overline{2w} = 1 - i - j - k$, direct multiplication gives $(2w)\, i\, \overline{2w} = 4j$, $(2w)\, j\, \overline{2w} = 4k$, $(2w)\, k\, \overline{2w} = 4i$, and $(2w)^3 = -8$; dividing by $N(2w) = 4$, Ad_w cycles $i \to j \to k \to i$ and $w^3 = -1$.

The automorphism Ad_w is the same for w and for $-w$, and $(-w)^3 = +1$; this section uses only the automorphism. Which of the two elements the drive is on spinors is settled in Corollary 7.8: it is $-w$.

The one place at which it bears on this paper is the number of steps in the time circle (Corollary 7.8).

This is the algebraic image of the corner three-cycle of Proposition 4.4, and it conjugates Ad_i to Ad_j to Ad_k, so that the three complex corners are carried with the three cycled points.

Fixed points of inversion. On T^n, $z \mapsto -z$ fixes the half-lattice $(\frac{1}{2}\mathbb{Z}/\mathbb{Z})^n$: two points on S^1, four on T^2, eight on T^3.

The Lefschetz number of $-I$ is $\det(I-(-I)) = 2^n$, in agreement.

Four corners, the number forced by the algebra, select $n = 2$.

The algebra at a point. P_1 is left in place by ι and by M_0, and its algebra is $\mathrm{Fix}(\mathrm{conj}) \cap \mathrm{Fix}(\mathrm{Ad}_w) = \mathbb{R}$; P_2, P_3, P_4 are left in place by ι and not by M_0, and their algebras are $\mathbb{C}_i, \mathbb{C}_j, \mathbb{C}_k$ (Appendix B); a point such as $(\frac{1}{3}, 0)$ is moved by ι to $(\frac{2}{3}, 0)$ and by M_0 to $(\frac{2}{3}, \frac{1}{3})$, is left in place by neither, and carries $\mathbb{H}$.

5.6 The pillowcase and its cones

Theorem 5.7 (the fibre)**.** *The quotient T^2/ι is the pillowcase orbifold: a sphere with four cone points of cone angle π, the unique closed Euclidean two-orbifold of signature $(2,2,2,2)$. Its orbifold Euler characteristic is zero and its Euler characteristic is two. [*R*]*

Proof. ι has order two, so each fixed point is a cone point of order two, angle $2\pi/2 = \pi$.

A closed two-orbifold with four cone points of order two on a sphere has orbifold Euler characteristic $2 - 4 \cdot \frac{1}{2} = 0$ (`L2_chi_orb_zero`), hence is Euclidean [19]; among the seventeen Euclidean two-orbifolds the one with signature $(2,2,2,2)$ and no mirror boundary is unique, $\mathbb{R}^2/p2 = T^2/\mathbb{Z}_2$.

If the underlying surface were not a sphere the characteristic would be negative.

For χ, two routes: the quotient formula $\chi(X/\sigma) = \frac{1}{2}(\chi(X) + \chi(\mathrm{Fix}\,\sigma)) = \frac{1}{2}(0+4) = 2$ [1]; and orbifold Gauss–Bonnet with the flat metric, $2\pi\chi = \int K + \sum_\alpha \mathrm{deficit}_\alpha = 0 + 4\pi$ [17], so $\chi = 2$ (`L2_chi_pillowcase`). □

The cone, by hand. Near a fixed point $z \mapsto -z$ is rotation by π about the point, so a small disc of angle 2π is folded to a cone of angle π with deficit π.

Four such deficits sum to $4\pi = 2\pi \cdot 2$, which is the Gauss–Bonnet count for $\chi = 2$ with a flat interior.

The alternative count, half the Euler characteristic of the torus plus half the number of fixed points, gives $\frac{1}{2}(0+4) = 2$ with no metric at all; that the two agree is the first consistency check between the metric and the combinatorial pictures of the fibre.

The four cone points are the geometry of Proposition 5.6: each is a point of angle deficit π at which the fibre algebra has collapsed to a commutative line, the deficit being the trace of the identification $z \sim -z$ that produced the line. In the covering torus with periods (L_ψ, L_ω) any two distinct corners are at distance at least $\frac{1}{2}\min(L_\psi, L_\omega)$: the corners are separated by the periods.

We return to the second axiom.

Proposition 5.8 (the geometry satisfies A2)**.** *Let s be the length along T and L the length of one step. Let v_1 be a unit eigenvector of M_0 with eigenvalue ϕ^2, let $v_2 = Sv_1$, where $S = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ is the quarter-turn, and let x, y be the coordinates on the fibre $T^2 = \mathbb{R}^2/\mathbb{Z}^2$ along v_1, v_2. Then v_2 is an eigenvector of M_0 with eigenvalue ϕ^{-2}. Let lengths on T together with the fibre be*

measured by $\mathrm{d}s^2 + e^{2\kappa s}\mathrm{d}x^2 + e^{-2\kappa s}\mathrm{d}y^2$*, the fibre at* $s + L$ *being joined to the fibre at* s *by the step. The lengths agree across the join if and only if* $e^{\kappa L} = \phi^2$*. The map* $(s, x, y) \mapsto (-s, -y, x)$*, time run backwards together with the quarter-turn, keeps these lengths, respects the join because* $SM_0S^{-1} = M_0^{-1}$*, and swaps* κ *for* $-\kappa$*. Applied twice it is* $(s, x, y) \mapsto (s, -x, -y)$*, the fold* ι*. The geometry therefore satisfies A2. [*D*]*

6 Dimension twelve

Theorem 6.1 (the dimension)**.** *Under Definition 2.7 a geometry satisfying A1–A5 has dimension divisible by four and at least twelve; under clause (c) of Definition 3.2 its dimension is twelve. [*D*]*

Proof. An almost-quaternionic structure makes each tangent space a module over the quaternions, so the real dimension is divisible by four: $4 \mid D$.

The time–space block $T \times \Sigma$ carries the quaternionic structure with the drive along T, so its dimension is a positive multiple of four.

Clause (c) takes the least total dimension, and a larger block only raises the total: the block has dimension four, that of $\mathbb{H}^\times = \mathbb{R}_{>0} \times S^3$ (Section 7 fixes Σ).

Hence $\dim F = D - 4$.

A3 places the full algebra with its four corners on F (Definition 2.7).

On a four-manifold every oriented Riemannian structure is almost-quaternionic, since $\mathrm{Sp}(1)\mathrm{Sp}(1) = SO(4)$ and the triple I, J, K is all of Λ^2_+: A3 would carry no information on F.

So F has quaternionic dimension at least two, $\dim F \geq 8$, $D \geq 12$ (`L3_dim_ge_twelve`).

Every D from four to eleven fails: either $4 \nmid D$, or $D \in \{4, 8\}$ with F of quaternionic dimension zero or one (`L3_below_twelve`); $D = 12$ is admissible (`L3_twelve_ok`); $D = 16, 20, \ldots$ are admissible and excluded by least dimension. □

Remark 6.2 (where the quaternionic structure carries information)*.* On a four-dimensional piece the almost-quaternionic structure of Definition 2.7 exists for any oriented metric, since $\mathrm{Sp}(1)\mathrm{Sp}(1) = SO(4)$ and I, J, K span Λ^2_+; it therefore carries no information there, and the time–space block $T \times \Sigma$ is such a piece. A single such structure on the eight-dimensional internal piece is not constructed here: grouping its factors as $\mathbb{CP}^2$ and $S^2 \times (T^2/\mathbb{Z}_2)$ and taking the triple of the preceding sentence on each, the two cannot be matched, since $\mathbb{CP}^2$ is not spin and $S^2 \times (T^2/\mathbb{Z}_2)$ is; the grouping is a choice, and that no grouping works is not shown. The fibre algebra itself is realised on the pillowcase factor of F, where it reduces to $\mathbb{R}, \mathbb{C}_i, \mathbb{C}_j, \mathbb{C}_k$ at the four corners and is $\mathbb{H}$ at every other point (Proposition 5.6 with Theorem 5.7; the four corners are the four cone points). The quaternionic structure therefore carries information in exactly one place, the pillowcase: everywhere else in $\mathcal{M}_{12}$ it is either automatic, on the four-dimensional blocks, or not constructed, on the eight-dimensional internal piece. That the structure exists on the whole of $\mathcal{M}_{12}$, which Theorem 6.1 requires of any geometry satisfying A1–A5, is assumed here and is not shown.

6.1 Worked cases

The elimination, dimension by dimension. The time–space block has dimension four throughout, so $\dim F = D - 4$ and the quaternionic dimension of F is $m = (D-4)/4$ when this is an integer.

D	$4 \mid D$?	$\dim F$	m	verdict
4	yes	0	0	no internal factor at all
5, 6, 7	no	–	–	no almost-quaternionic structure on M
8	yes	4	1	$\mathrm{Sp}(1)\mathrm{Sp}(1) = SO(4)$: A3 vacuous on F
9, 10, 11	no	–	–	no almost-quaternionic structure on M
12	yes	8	2	admissible; least
13, 14, 15	no	–	–	no almost-quaternionic structure on M
16	yes	12	3	admissible; excluded by least dimension

The case $D = 8$ deserves a sentence, because non-commutativity alone does not exclude it: $\mathbb{H}(2)$, the quaternionic 2×2 matrices that would act in quaternionic dimension two on the whole of M, is non-commutative, and an eight-manifold can carry an almost-quaternionic structure. What fails is the placement of the algebra on the internal factor. With four dimensions spent on the time–space block, the internal factor of an eight-dimensional geometry is a four-manifold, and on a four-manifold the three almost-complex structures I, J, K span the self-dual two-forms of any oriented metric: the quaternionic structure is automatic, carries no information, and the four corners of Section 5 have nowhere to live. Twelve is the first dimension at which the internal factor is large enough for A3 to say something about it.

The split within twelve: $\dim \Sigma + \dim(\text{base}) = 9$, $\dim \Sigma \in \{3, 7\}$ from $\dim \Sigma + 1 \equiv 0$, $\dim(\text{base}) \in \{2, 6\}$ from $\dim(\text{base}) + 2 \equiv 0 \pmod 4$. The pairs summing to nine are $(3, 6)$ and $(7, 2)$. The second makes F four-dimensional and fails as $D = 8$ did. So $\dim \Sigma = 3$, $\dim(\text{base}) = 6$, $\dim F = 8$.

Remark 6.3 (the split). Within $D = 12$, $\dim \Sigma + \dim(\text{base}) = 9$ with $T \times \Sigma$ and $F = \text{base} \times T^2/\mathbb{Z}_2$ both quaternionic forces $(\dim \Sigma + 1, \dim(\text{base}) + 2) \equiv (0, 0) \pmod 4$. The congruences alone admit $(3, 6)$ and $(7, 2)$; the second gives F four-dimensional, excluded by the same argument as $D = 8$. With that condition the split is uniquely $(3, 6)$ (`L3_split_three_six`; the excluded alternative is recorded in `L3_seven_two_meets_congruences`). The certificate surfaced the second solution; the condition excluding it is stated rather than assumed.

7 The base

This section takes up the base, under six headings: the sphere of imaginary units, the space of complex directions, the twist that is necessary, the twist of the fibre, worked cases, and the spatial factor.

7.1 The sphere of imaginary units

Proposition 7.1 (the two-sphere)**.** *The elements $u \in \mathrm{Im}\,\mathbb{H}$ with $u^2 = -1$ form a two-sphere, on which the inner automorphisms Ad_g, $g \in \mathrm{Sp}(1)$, act transitively through $\mathrm{Sp}(1) \to SO(3)$. If one unit i is selected, its stabiliser is $U(1)$ and the orbit of i is $\mathrm{Sp}(1)/U(1) = S^2$. Under Definition 2.8 this sphere is a factor of the base. [D]*

Proof. $\mathrm{Im}\,\mathbb{H} \cong \mathbb{R}^3$ and $u^2 = -|u|^2$, so the condition is $|u| = 1$.

The three corner units are an orthonormal frame, the integer points of the sphere (`L4_unit_sphere_axes`); V acts on it as the rotations by π about the three axes.

The stabiliser of i in $\mathrm{Sp}(1)$ is $\{e^{i\theta}\}$, and $\mathrm{Sp}(1)/U(1)$ is the Hopf quotient $S^3 \to S^2$.

The last sentence is Definition 2.8. □

7.2 The space of complex directions

Proposition 7.2 (the projective plane)**.** *Let the three imaginary units be complex-valued (the tenth choice), so that the corner data take values in $\mathbb{C} \otimes \mathrm{Im}\,\mathbb{H} \cong \mathbb{C}^3$, with $z_1 i + z_2 j + z_3 k$ of length $(|z_1|^2 + |z_2|^2 + |z_3|^2)^{1/2}$, and let $U(3)$ be the group of complex-linear maps of $\mathbb{C}^3$ that keep this length. The geometry records the direction of a complex-valued unit and not the unit: a non-zero element of $\mathbb{C}^3$ taken up to multiplication by a non-zero complex number, that is, a complex line in $\mathbb{C}^3$. That its size is not recorded is the absence of an overall scale (Section 3); that its phase is not recorded is a choice and is not derived. The space of complex lines in $\mathbb{C}^3$ is $\mathbb{CP}^2 = U(3)/(U(1) \times U(2)) = SU(3)/U(2)$; its Euler characteristic three counts the units, its tangent bundle has $c_1 = 3h$, and its anticanonical line is $\mathcal{O}(3)$. Under Definition 2.8 it is a factor of the base. [D]*

Proof. The three coordinate lines [1:0:0], [0:1:0], [0:0:1] are the units; the maximal torus of $U(3)$ fixes exactly these, and the Lefschetz count gives $\chi = 3$ (`L4_chi_counts`).

By the Euler sequence $0 \to \mathcal{O} \to \mathcal{O}(1)^{\oplus 3} \to T\mathbb{CP}^2 \to 0$ and the Whitney formula, $c(T\mathbb{CP}^2) = (1+h)^3 = 1 + 3h + 3h^2$ in $\mathbb{Z}[h]/(h^3)$ [6] (`L4_chern_CP2`); $\int c_2 = 3 = \chi$ checks the smallest case.

This proposition is the least direct of the section and is tagged accordingly. □

7.3 The twist is necessary

Proposition 7.3 (Spin^c is forced)**.** *Under Definition 2.7 a spinor bundle exists on M. Since $w_2(\mathbb{CP}^2) = h \bmod 2 \neq 0$, no spin structure exists on a geometry whose base contains $\mathbb{CP}^2$, and spinors exist only through a Spin^c structure, whose determinant line satisfies $c_1(L) \equiv w_2 \pmod 2$ and is therefore odd. A Spin^c structure exists, and clause (b) of Definition 3.2 admits no structure group larger than is needed. The structure induced by the almost-complex structure of A3's selected unit is the canonical one, with $L = K^{-1} = \mathcal{O}(3)$, and every other admissible line is $\mathcal{O}(3) \otimes E^2$ for a line E, excluded by clause (b): the twisting line is an object that nothing requires. [D]*

Proof. An oriented manifold carries spinors iff $w_2 = 0$ or w_2 lifts to an integral class, the latter being the Spin^c condition with $c_1(L) \equiv w_2$ [10].

$w_2(\mathbb{CP}^2)$ is the reduction of $c_1 = 3h$, which is $h \neq 0$ (`L4_not_spin_and_congruence`); $3h$ is integral, so Spin^c exists and $c_1(L)$ is odd.

A Spin^c structure therefore suffices.

A larger structure group, such as that of a Spin^h structure, is not needed, and clause (b) does not admit it.

Every almost-complex manifold carries a canonical Spin^c structure with determinant K^{-1} [10, 2]; on $\mathbb{CP}^2$, $K^{-1} = \mathcal{O}(3)$; the Spin^c structures form a torsor over $H^2(\mathbb{Z})$ by $L \mapsto L \otimes E^2$, and $\mathcal{O}(1), \mathcal{O}(5)$ are $\mathcal{O}(3) \otimes \mathcal{O}(\mp 1)^2$ (`L4_odd_lines`).

The S^2 factor is spin and the pillowcase is flat, so neither obstructs.

The flux parity, Q_c odd on $\mathbb{CP}^2$ and Q_w even on S^2, is the same congruence on the flux side (`L4_flux_parity`). □

7.4 The twist of the fibre

Proposition 7.3 fixes the determinant line on $\mathbb{CP}^2$ and says nothing of the fibre. The second cohomology of $\mathbb{CP}^2 \times S^2 \times T^2$ is one integer for each factor, with no mixed term, because the first cohomology of $\mathbb{CP}^2 \times S^2$ vanishes [7]. The determinant line of a Spin^c structure therefore carries three independent integers, odd on $\mathbb{CP}^2$ and even on the other two, and the canonical structure has zero on the torus. On S^2 the canonical structure has the integer two. The sphere is the complex projective line, and its anticanonical line is its tangent line; in the coordinate $w = 1/z$ the vector field $\partial/\partial z$ is $-w^2\,\partial/\partial w$, which has one zero, of order two, and no pole. Zero or four would need a twisting line, which clause (b) of Definition 3.2 removes. A twist of the fibre by a line of degree d on the torus multiplies the determinant line by the square of that line, so that the integer on the torus becomes $2d$. Whether the fibre is twisted, and by how much, is decided by the fifth axiom.

Let E be the covering torus of the pillowcase with a complex structure for which the fold is $\iota(z) = -z$, and let $P_1, \ldots, P_4$ be the four corners. The counts below do not depend on which such complex structure is taken.

Definition 7.4 (flux and states)**.** A *twist of the fibre* of *flux* $d \geq 1$ is the line $\mathcal{O}(D)$ on E of a divisor $D = \sum a_i P_i$ on the corners, with $a_i \geq 0$ and $\sum a_i = d$. A twist of flux zero is a flat line with holonomy ± 1: the trivial one, or one of the three *half-turn twists*.

A *state at rest* is a holomorphic section of the twist: for $d \geq 1$, a function f on the covering torus E whose poles are bounded by D. The fold acts on states at rest by $f(z) \mapsto f(-z)$ or by $f(z) \mapsto -f(-z)$, its two lifts, called here the first and the second. A *state on the pillowcase*, for a given lift, is a state at rest that the lift leaves unchanged. The torus E is the fibre before it is folded, and the pillowcase is the fibre folded (Theorem 5.7); for a given lift, the states of the folded fibre at rest are the states on the pillowcase for that lift, and a state at rest on E that the lift does not leave unchanged is not one of them. Which lift is called the first depends on the divisor chosen for a line, and the statements below speak of one lift or the other.

The *corner sign* of the twist at P_i is $(-1)^{a_i}$, the action of the first lift on the fibre of $\mathcal{O}(D)$ there. The four signs are taken up to one overall sign. A corner is *odd* if its sign differs from that of the other three.

The *structure* of a twist is its flux together with its corner signs up to the overall sign or, for flux zero, its holonomy. It is this that clause (ii) of Definition 2.1 speaks of. The drive is not a holomorphic map of E; it acts on structures through its action on the corners.

Two states on the pillowcase, for the same lift of the fold, have the same shadow: they differ only in the internal piece. The states at rest of Definition 2.1(iii) are states on the pillowcase for one of the two lifts of the fold. This is the reading of Definition 2.1(iii).

For the canonical Spin^c structure the states at rest are the harmonic spinors of the fibre. We use this as motivation for the definition of a state, and the counts below do not depend on it. One fact about spinors is used, in Corollary 7.7: on the untwisted spinors of the fibre, the functions and the $(0,1)$-forms on E, the fold acts by $+1$ and by -1 at every corner alike, since $\iota^* \mathrm{d}\bar{z} = -\mathrm{d}\bar{z}$. The corner signs of the spinors, up to the overall sign, are therefore those of the twist.

Lemma 7.5 (states, the fold and the drive)**.** *Let D be a divisor of degree $d \geq 1$ on the corners. (a) There are exactly d linearly independent states. (b) Under the first lift the number of states on the pillowcase is $1 + \sum_i \lfloor a_i/2 \rfloor$; under the second it is d less that number. (c) The corner signs are carried to themselves by one step of the drive if and only if the four are alike or the odd one is at P_1; otherwise they return after three steps and not before. (d) A half-turn twist has no state at rest, and the drive carries the three half-turn twists round a cycle of three. [*R*]*

Proof. (b) An even elliptic function is a rational function of the Weierstrass function $\wp$ [20], that is, a function on the quotient sphere E/ι, and its order at a corner is twice its order at the image point.

An even state is therefore a function on the sphere with a pole of order at most $\lfloor a_i/2 \rfloor$ at the image of P_i.

Such functions form a space of dimension $1 + \sum_i \lfloor a_i/2 \rfloor$, by partial fractions.

(a) The first lift is an involution of the states, so every state is the sum of an even state and an odd one; the second lift keeps the odd ones.

An odd state is $\wp'$ times an even function g, and $\wp'$ has a triple pole at P_1 and a simple zero at each of P_2, P_3, P_4 [20].

So g is a function on the sphere with a pole of order at most $\lceil a_i/2 \rceil$ at the image of P_i for $i = 2, 3, 4$, and of order at most $\lceil a_1/2 \rceil - 2$ at the image of P_1, a negative bound meaning a zero of at least that order.

Such functions form a space of dimension $\sum_i \lceil a_i/2 \rceil - 1$.

The two dimensions add up to $\sum_i a_i = d$.

(c) The drive fixes P_1 and cycles $P_2 \to P_3 \to P_4$ (Proposition 4.4).

A pattern with all four signs alike, or with the odd sign at P_1, is unchanged.

A pattern with one odd sign at a cycled corner, or with two signs of each kind, is moved, and returns when the corners do (`L6_fixed_patterns`, `L6_return_three`).

(d) A section of a line of degree zero has no zeros, so it trivialises the line; a half-turn twist is not trivial, its holonomy being -1 on some loop, so it has no section.

The three half-turn twists are the three non-trivial characters of the lattice modulo two, which the drive permutes as it permutes the three cycled corners. □

Proposition 7.6 (the twist of the fibre)**.** *Let the fibre satisfy clauses (ii) and (iii) of Definition 2.1, the states of clause (iii) being states on the pillowcase for one of the two lifts of the fold, and let the flux be the least that does so. Then the flux is three. The corner signs are three alike and one odd, and the odd corner is one of the three that the drive cycles. For that lift the fibre has two states on the pillowcase, and the fold removes a third.*

*Taken singly, clause (ii) is met with least flux by a half-turn twist of flux zero, for which the fibre has no state at rest; and clause (iii) is met with least flux by flux two with the four corner signs alike, a structure the drive leaves in place. [*D*]*

Proof. Flux zero.

The trivial line is carried to itself by the drive and fails clause (ii).

A half-turn twist meets clause (ii) and has no state (Lemma 7.5(d)), so it fails clause (iii).

Flux one.

There is one state (Lemma 7.5(a)).

Clause (iii) fails (`L6_flux_one`).

Flux two.

For $D = 2P_i$ the first lift keeps two states and the corner signs are alike.

For $D = P_i + P_j$ with $i \neq j$ each lift keeps one state (Lemma 7.5(b)).

Clause (iii) therefore forces $D = 2P_i$, whose structure, with the four corner signs alike, the drive leaves in place (Lemma 7.5(c)).

Clause (ii) fails (`L6_flux_two`).

Flux three.

For $D = 3P_i$ and for $D = 2P_i + P_j$ the first lift keeps two states and removes one; for $D = P_i + P_j + P_k$ the second lift does.

Clause (iii) holds in every case.

The corner signs have one odd corner: P_i, P_j and the fourth corner respectively.

Clause (ii) holds exactly when the odd corner is not P_1 (Lemma 7.5(c)).

The case analysis is certified, taking the count of Lemma 7.5(b) as given (`L6_flux_three_states`, `L6_flux_three_one_odd`, `L6_flux_three_cycled`, `L6_least_flux`).

The single clauses are the cases of flux zero and flux two above. □

The smallest case, by hand. Take $D = 3P_4$, and write $\wp$ for the Weierstrass function of E and $e_4 = \wp(P_4)$.

The three states are 1, $1/(\wp - e_4)$ and $\wp'/(\wp - e_4)^2$.

The first two are even and the third is odd.

The corner signs are $(+,+,+,-)$ (`L6_smallest_case`).

Three remarks. That the least flux is taken is clause (c) of Definition 3.2. The twisting line that clause (b) removes in Proposition 7.3 is a line on $\mathbb{CP}^2$ that nothing forces; the twist of the fibre is forced by the fifth axiom, and clause (c) takes the least of it. And the flux three found here and the three of $c_1(\mathbb{CP}^2)$ are the same integer, but this paper does not derive one from the other. There may be a relationship, but that has not been found in this paper.

Corollary 7.7 (the corner lift)**.** *With the twist of Proposition 7.6 the lift of ι to the spinors of the fibre acts at the four corners by signs whose product is -1: one corner carries the sign opposite to that of the other three, and it is one of the three corners cycled by the drive. Naming the corners so that it is P_4, the sign at $C_{m,n} = (\frac{m}{2}, \frac{n}{2})$ is $(-1)^{mn}$, up to one overall sign. For an even flux the product would be $+1$. [D]*

Proof. The product of the corner signs of $\mathcal{O}(D)$ is $\prod_i (-1)^{a_i} = (-1)^d$ (`L6_corner_product`), and the untwisted spinors contribute the same sign at every corner.

The rest is Proposition 7.6.

The eight functions $(\mathbb{Z}_2)^2 \to \mathbb{Z}_2$ vanishing at the origin split into the four characters, with product $+1$, and the four quadratic refinements of the intersection pairing, with product -1.

An even flux gives a character and an odd flux a refinement.

The sign table is `L4_four_corner_product`. □

That the odd corner is a cycled one is clause (ii) of the fifth axiom at work. With the odd corner at P_1 the drive would leave the structure in place, and one thing would have one shadow.

Among the three cycled corners there is nothing further to choose. The three patterns are one pattern at three successive steps, and to name one of them is to say which step is called the first. The pattern $(-1)^{mn}$ is the one of the three that the quarter-turn S, as written in Theorem 4.2, leaves in place: S fixes P_1 and P_4 and exchanges P_2 and P_3. The quarter-turns $M_0 S M_0^{-1}$ and $M_0^{-1} S M_0$ leave the other two patterns in place in the same way, which is the same statement: nothing distinguishes the three but the step at which one starts.

Corollary 7.8 (the time circle)**.** *With the corner signs of Corollary 7.7, the signs are carried to themselves by n steps of the drive if and only if three divides n. The lifted fold being one structure on the whole geometry, the time circle T is a whole number of steps divisible by three. We take the least, three. On spinors the drive is then the element $-w$, of cube $+1$, and not w, of cube -1, for which six steps would be needed: clauses (b) and (c) of Definition 3.2 select it. Once round T the fibre is then glued to itself by $M_0^3 = \left(\begin{smallmatrix} 13 & 8 \\ 8 & 5 \end{smallmatrix}\right)$, of trace eighteen, which leaves every corner in place. [D]*

Proof. One step moves the odd corner from P_4 to P_2, a second from P_2 to P_3, a third from P_3 to P_4 (Proposition 4.4).

The reduction of M_0 modulo two has order three (`L1_M0_mod2_order3`), so the pattern returns after three steps and not before.

Time is a circle (Definition 2.3), and the time factor is the torus bundle whose step is M_0 (Theorem 4.2): once round T the fibre is glued to itself by a power of the drive, the number of steps in T.

A structure defined on the whole geometry, carried once round the time circle, returns to itself; that the lifted fold is such a structure is the hypothesis of the statement.

The number of steps in T is therefore a multiple of three.

$M_0^3 = F^6$ has the entries $(13, 8, 8, 5)$ (`L1_Fib_powers`) and is the identity modulo two.

What is forced is divisibility by three.

That the least multiple is taken is clause (c) of Definition 3.2.

It remains to say which of the two elements w and $-w$ of Section 5.5 the drive is on spinors.

With w, three steps return the corner signs with the overall sign reversed, since $w^3 = -1$, and six steps are needed; with $-w$ three suffice, since $(-w)^3 = +1$.

The first five axioms allow both.

Clause (c) takes the least number of steps, which is three, and with it $-w$.

Clause (b) makes the same selection when every symmetry of the torus, and not only the fold and the drive, is given an element; that every symmetry is given one is a hypothesis of this paragraph.

The group $\mathrm{SL}_2(\mathbb{Z})$ acts on the upper half-plane, the stabiliser of the point i is cyclic of order four and that of $e^{\pi i/3}$ cyclic of order six, and $\mathrm{SL}_2(\mathbb{Z})$ is the amalgam of the two over their common subgroup of order two [16].

The quarter-turn S fixes i and has order four; $U = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 1 \end{smallmatrix}\right)$ sends z to $1/(1-z)$, fixes $e^{\pi i/3}$ and has order six; and $S^2 = U^3 = -1$.

An assignment of unit quaternions to the symmetries of the torus is therefore a pair (s, u) with $s^4 = 1$ and $s^2 = u^3$, and it gives the drive the element $d = u^{-1}s^{-1}u^{-2}s^{-1}$, since $M_0 = U^{-1}S^{-1}U^{-2}S^{-1}$.

We ask that the assignment move the three lines $\mathbb{C}_i, \mathbb{C}_j, \mathbb{C}_k$ as the symmetry moves the corners P_2, P_3, P_4 (Proposition 5.6), with no condition on signs.

S exchanges P_2 and P_3, so s is one of the four elements $(\pm i \pm j)/\sqrt{2}$, and $s^2 = -1$.

U cycles the three corners, so u covers one of the four rotations of order three that cycle the three lines, and of the two elements covering a rotation one has cube -1: there are four choices of u, of the form $\frac{1}{2}(1 \pm i \pm j \pm k)$.

There are thus sixteen assignments, all in the binary octahedral group of order forty-eight.

Direct multiplication, carried out exactly in $\mathbb{Q}(\sqrt{2})$ and repeated independently, gives two families of eight.

In the first $d^3 = +1$, and s and u generate a group of order twelve; a symmetry that leaves every corner in place then leaves i, j, k in place.

In the second $d^3 = -1$, and s and u generate the whole group of order forty-eight.

The larger group is an object that nothing requires, and clause (b) removes it.

Six steps is thus a consistent alternative that the sixth axiom declines, and not an impossibility. □

Two things follow. First, the numbers attached to the drive in Section 4 are numbers per step: the stretch is ϕ^2 and the entropy $2\log\phi$ per step, and they are ϕ^6 and $6\log\phi$ once round the time circle.

Second, the entropy of clause (c) is that of the step and not of the map once round the time circle. The map once round is a power of the step and is not chosen separately. That the clause is applied to the step is a choice of what the clause is applied to, and is recorded as such.

7.5 Worked cases

The unit sphere and its axes. $u = bi + cj + dk$ has $u^2 = -(b^2 + c^2 + d^2)$, so $u^2 = -1$ is the unit sphere in $\mathbb{R}^3$.

Its integer points are $\pm i, \pm j, \pm k$: the corner units, an orthonormal frame.

Ad_i fixes $\pm i$ and sends $j \mapsto -j$, $k \mapsto -k$: rotation by π about the i-axis; the Klein four-group V is the group of such rotations about the three axes.

The circle $U(1) = \{\cos\theta + i\sin\theta\}$ fixes i and rotates the equator $\{cj + dk\}$, the units anticommuting with i, among which are j and k.

The Euler sequence. From the exact sequence

$$0 \to \mathcal{O} \to \mathcal{O}(1)^{\oplus 3} \to T\mathbb{CP}^2 \to 0$$

and $c(\mathcal{O}) = 1$, $c(\mathcal{O}(1)) = 1 + h$, the Whitney formula gives

$$c(T\mathbb{CP}^2) = (1 + h)^3 = 1 + 3h + 3h^2 + h^3,$$

and $h^3 = 0$ on $\mathbb{CP}^2$. So $c_1 = 3h$, $c_2 = 3h^2$, $\int c_2 = 3 = \chi(\mathbb{CP}^2)$, the Betti numbers being $1, 0, 1, 0, 1$. The anticanonical line is $\det T^{1,0}$ with $c_1 = 3h$, i.e. $\mathcal{O}(3)$; the canonical line is $\mathcal{O}(-3)$.

The admissible lines. $c_1(L) \equiv w_2 = h \pmod 2$ admits $L = \mathcal{O}(n)$ for every odd n.

The canonical structure has $L = K^{-1} = \mathcal{O}(3)$ and no further datum.

Its neighbours are twists of it,

$$\mathcal{O}(1) = \mathcal{O}(3) \otimes \mathcal{O}(-1)^2, \qquad \mathcal{O}(5) = \mathcal{O}(3) \otimes \mathcal{O}(1)^2,$$

each requiring a twisting line $E = \mathcal{O}(\mp 1)$: a further object, which clause (b) removes.

The eight corner functions. A function $q : \{0,1\}^2 \to \mathbb{Z}_2$ with $q(0,0) = 0$ is one of eight.

The four characters $q = am + bn$ have $\sum q = 2(a+b) \equiv 0$ and corner product $+1$; the four quadratic refinements $q = mn + am + bn$ have $\sum q \equiv 1$ and product -1.

For $q = mn$: values $0, 0, 0, 1$ at $(0,0), (1,0), (0,1), (1,1)$, signs $+, +, +, -$.

Three corners carry $+1$ and one carries -1, and it is the odd flux of the fibre that puts the lift in the refinement class (Proposition 7.6).

Flux parity. The flux quantum through a closed surface S is $Q_S = \int_S c_1(L) \equiv \langle w_2, [S] \rangle$ mod two.

Through the generating $\mathbb{CP}^1 \subset \mathbb{CP}^2$, $Q_c = 3h \cdot [\mathbb{CP}^1] = 3$, odd.

Through S^2, which is spin, Q_w is even.

7.6 The spatial factor

Proposition 7.9 ($\Sigma = S^3$)**.** *The spatial piece Σ of the four-dimensional block of Theorem 6.1 is a compact connected three-dimensional manifold without boundary (Definition 2.3). Clause (b) of Definition 3.2 admits no class of loops that nothing requires, and the loop of Definition 2.3 is supplied by the time circle: the fundamental group of Σ is trivial; and a closed, simply connected three-manifold is the three-sphere [12]. So $\Sigma = S^3 = \mathrm{Sp}(1)$, the unit quaternions, and the block is modelled on $\mathbb{H}^\times = \mathbb{R}_{>0} \times S^3$. [D]*

The quotients of the sphere by $\mathbb{Z}_2$ and $\mathbb{Z}_3$ have no modulus and are removed only by clause (b), which is recorded as the reason: their loops are topology that no axiom requires.

Theorem 7.10 (the base)**.** *The base of a minimal geometry is $\mathbb{CP}^2 \times S^2$, its Spin^c determinant line is $\mathcal{O}(3)$, its spatial factor is S^3, and the Euler characteristic of the internal factor is $\chi(K_8) = \chi(\mathbb{CP}^2)\,\chi(S^2)\,\chi(T^2/\mathbb{Z}_2) = 3 \cdot 2 \cdot 2 = 12$. [D]*

Proof. Propositions 7.1, 7.2, 7.3, 7.9 supply the factors.

Definition 2.8 calls for two factors of the base, S^2 (Proposition 7.1) and $\mathbb{CP}^2$ (Proposition 7.2), of dimensions two and four.

Clause (b) of Definition 3.2 admits no further factor: it would be an object that A1–A5 do not require.

The base is therefore $\mathbb{CP}^2 \times S^2$, of dimension six, the dimension that Remark 6.3 leaves for it.

The warping is a metric datum and does not change the homotopy type, so χ is multiplicative by Künneth [7], with $\chi(T^2/\mathbb{Z}_2) = 2$ from Theorem 5.7 (`L3_chi_K8`).

The identity $12 = 1 + 3 + 8 = \dim \mathcal{M}_{12}$ holds and is recorded, not used. □

8 The minimality theorem

8.1 Statement and proof

The theorem collects the four sections into one statement. Its hypotheses are the six axioms, the readings of Section 2 and the three of Section 3, and one declared input; its conclusion names every factor and every discrete datum of the geometry. What a referee should check is that no factor is named before it is forced, and the proof is written so that each clause points to the section that forces it.

Theorem 8.1 (minimality)**.** *Among geometries $(T\times\Sigma)\times F$ satisfying A1–A5 under the readings of Definitions 2.1, 2.3, 2.4(i), 2.7 and 2.8 and the tenth choice, with the continuous data of the geometry that this paper does not derive held fixed, which is its one declared input, the geometry that satisfies A6 in the sense of Definition 3.2, the minimal geometry, is unique and is*

$$\mathcal{M}_{12} = (S^1_{\mathrm{Sol}}\times S^3)\times\left[(\mathbb{CP}^2\times S^2)\times_w (T^2/\mathbb{Z}_2)\right]^{Spin^c},$$

with step $\left(\begin{smallmatrix}2&1\\1&1\end{smallmatrix}\right)$ and a time circle of three steps, flux three on the fibre with one odd corner (Proposition 7.6), determinant line $\mathcal{O}(3)$, the integer two on S^2, $\chi(K_8)=12$ and dimension twelve. The primes two, three and five of its topological integers are outputs of the selection.

Proof. A5 gives the family $(T\times\Sigma)\times F$ in which the other four axioms are stated (Definition 2.1) and, through its clauses (ii) and (iii), the twist of the fibre and the corner signs (Proposition 7.6, Corollary 7.7).

A4 with clause (a) gives the drive, and clause (c) agrees (Theorem 4.2, Proposition 4.5).

A1–A4 with clause (c) give the fibre, its algebra, its corners and the pillowcase (Section 5).

A3 with clause (c) gives dimension twelve and the split $(3,6)$ (Theorem 6.1, Remark 6.3).

A3 with Definition 2.8 gives S^2 and $\mathbb{CP}^2$, with Spin^c necessary and $L=\mathcal{O}(3)$ and $\Sigma=S^3$ by clause (b) (Theorem 7.10).

Corollaries 7.7 and 7.8 give the time circle, of the least number of steps by clause (c), as the least flux of the twist is.

The primes: two enters through the involution of A2 and $\chi(S^2)$; three through the three imaginary units, $\chi(\mathbb{CP}^2)=3=c_1$; five through ϕ on the drive alone.

No clause of Definition 3.2 speaks of these primes: that the topological integers lie in $\{2^a3^b\}$, the prime five entering only through the drive, is an output. □

The deflationary reading. The fifth axiom enters the theorem twice: as the division of the shape into the part that is seen and the part that is not, and, through its two further clauses, as the twist of the fibre. The second rests on a reading, that an object is a state of the internal piece at rest and that its shadow is what is left when the internal piece is forgotten. A reader who does not take that reading keeps the division and loses Proposition 7.6 and Corollaries 7.7 and 7.8. The theorem asserts uniqueness among minimal realisations under the stated readings; it does not assert that a non-minimal geometry, one of dimension sixteen or with a lens-space

spatial factor, fails the first five axioms. That the continuous data of the geometry that this paper does not derive are held fixed is the one declared input of the theorem; it would fail if a continuous datum of the geometry that this paper does not derive were not held fixed, and the geometry would then not meet clause (a). The overall size is not an input: under the sixth axiom it has no meaning. The three clauses of Definition 3.2 are readings of the sixth axiom and its words do not force them: a reader who does not take clause (c) has dimension twelve, the torus T^2 as the fibre of F carrying the drive, flux three and three steps as the least admissible values and not as the only ones, and a reader who allows relabellings that reverse orientation has the drive by clause (c) alone. The readings are definitions, not results: a reader who takes Definition 2.8 in its fibrewise form, under which the sphere of units exists over each point but is not a factor, will not obtain Propositions 7.1 and 7.2, and the base is then determined among products only. Proposition 7.2 is the least direct step and is tagged [D].

8.2 The elimination table

Competitor	Fails at	Where	Tag
non-warped product	A2(i)	Definition 2.4	[D]
$D \not\equiv 0$ (4)	no quaternionic structure	Theorem 6.1	[D]
$D = 4$, $D = 8$	F of quaternionic dimension < 2	Theorem 6.1	[D]
$D = 16, 20, \ldots$	least dimension	Definition 3.2(c)	[D]
trace 2	entropy zero	Proposition 4.1	[R]
trace ≥ 4	several drives; least entropy	Thm. 4.2, Prop. 4.5	[R]
$M_2(\mathbb{R})$ as fibre algebra	non-compact unit groups	Proposition 5.2	[D]
T^n, $n \neq 2$, as fibre	$2^n \neq 4$ corners	Proposition 5.5	[D]
$T^4/\mathbb{Z}_2$, K3 as internal factor	16 and 0 corners; $D = 8$	Prop. 5.3, Thm. 6.1	[D]
$S^3/\mathbb{Z}_2$, $S^3/\mathbb{Z}_3$	loops that nothing requires	Proposition 7.9	[D]
spin bases: $S^2 \times S^2 \times S^2$, $\mathbb{CP}^3$, S^6	no $\mathbb{CP}^2$ factor	Definition 2.8, Definition 3.2(b)	[D]
other six-dimensional bases: $\mathbb{CP}^2 \times T^2$, T^6	no S^2 factor, and for T^6 no $\mathbb{CP}^2$ either	Definition 2.8, Definition 3.2(b)	[D]
irreducible bases: the quadric Q^3	no $\mathrm{Sp}(1)/U(1)$ factor	Proposition 7.1	[D]
$\mathcal{O}(1)$, $\mathcal{O}(5)$	unforced twisting line	Proposition 7.3	[D]

8.3 Cross-check

Witten [21] observes that seven is the least dimension of a manifold with $SU(3) \times SU(2) \times U(1)$ symmetry, and that $\mathbb{CP}^2 \times S^2 \times S^1 = SU(3)/U(2) \times SU(2)/U(1) \times U(1)$ is one of the seven-dimensional manifolds that have it. Propositions 7.1 and 7.2 produce $\mathrm{Sp}(1)/U(1)$ and $SU(3)/U(2)$ from the three imaginary units with no reference to a gauge group. The two routes are logically independent and land on the same base; Witten's result is used only as this confirmation, never as a premise.

9 Applications and outlook

The geometry (1) is now a theorem of six axioms, and the derivation fixes more than the manifold. The four corners of the pillowcase are cone points of angle π, each the locus where the quaternionic algebra has collapsed to a commutative line, with the drive fixing one and cycling the other three (Propositions 5.6, 4.4); the determinant line is odd; the fibre carries three units of flux, with two states on the pillowcase and one removed by the fold (Proposition 7.6); the corner lift has product -1 (Corollary 7.7); the time circle is three steps of the drive (Corollary 7.8). These are the data that the papers of the programme consume in deriving the lepton and quark mixing structure from cone deficits and corner walks, and the present paper fixes their origin. We do not pursue those consequences here. [M]

The sequel derives the field equation of the programme, with its coupling to the Coherent Field, from the same axioms on the geometry derived here; the two sheets of the involution of A2 and the non-commutative structure of A3 are the variables in which that equation is written.

The sixth axiom gives that nothing is absolute: there is no ruler outside the one thing. The fifth gives that the view from inside is incomplete: what is seen does not determine what is there (Remark 2.2). Together they resemble Gödel's first incompleteness theorem, that a consistent formal system in which a certain amount of arithmetic can be carried out leaves statements it can neither prove nor disprove [5]. The theorem is about formal systems and is not applied here. The resemblance is recorded here and nothing is derived from it. [M]

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A Machine-verified certificates

Six files, core library only, no Mathlib, no author axioms, zero `sorry`; sixty-six theorems. The sixth file, `M1_L7_onedrive.lean`, was written for Lean 4.29.1, the toolchain of the author's Lean project, uses its core tactics `grind` and `push_cast`, and was compiled on that toolchain only. The first five, with fifty-eight theorems, were written for Lean 4.15.0 and were compiled independently on two toolchains, the core toolchain used in drafting and the author's Lean project, with identical results; no proof of theirs depends on the default simplifier set of a particular Lean version. Classical theorems (Frobenius, Thurston, Latimer–MacDuffee, Lawson–Michelsohn, Salamon) are named in file headers and are never Lean axioms; the certificates establish the finite and arithmetic content those theorems consume.

`M1_L1_drive.lean` (Section 4): $M_0 \in \mathrm{SL}_2(\mathbb{Z})$ of trace three; $SM_0S^{-1} = M_0^{-1}$; $t^2 > 4$, $t > 0 \Rightarrow t \geq 3$; reduced forms of discriminant five are $(\pm 1, 1, \mp 1)$, one cycle; odd trace $\Leftrightarrow$ order three in $\mathrm{SL}_2(\mathbb{F}_2)$, by finite check; $\bar{M}_0$ of order three; $M_0 = F^2$ with F the Fibonacci matrix, the Fibonacci recursion in the entries, and $F \notin \mathrm{SL}_2(\mathbb{Z})$; $\det(M_0 - I) = -1$ (not used in the text).

`M1_L2_fibre.lean` (Section 5): Hamilton's relations; $ij \neq ji$; $N(q) = q\bar{q}$, positive-definite; fixed sets of conjugation and of $\mathrm{Ad}_i, \mathrm{Ad}_j, \mathrm{Ad}_k$; $\iota \circ \mathrm{Ad}_i$ fixes $i^{\perp}$, not closed; Ad_w cycles $i \to j \to k$, $(2w)^3 = -8$; corner count four; $2^n = 4 \Rightarrow n = 2$; $\chi_{\mathrm{orb}} = 0$; $\chi = 2$.

`M1_L3_dimension.lean` (Section 6): $D = 4 + 4m$, $m \geq 2 \Rightarrow D \geq 12$; dimensions four to eleven fail; twelve admissible; split $(3, 6)$ unique given $\dim F \geq 8$, and $(7, 2)$ recorded as the congruence solution excluded; $3 \cdot 2 \cdot 2 = 12 = 1 + 3 + 8$.

`M1_L4_base.lean` (Section 7): integer points of the unit sphere of $\mathrm{Im}\,\mathbb{H}$ are the three axes; $(1 + h)^3$ coefficients; $3 \bmod 2 = 1$ and the congruence; odd lines $3 + 2e$; flux parity; four-corner product -1; $\chi(\mathbb{CP}^2) = 3$ as a count.

`M1_L6_twist.lean` (Section 7.4): for the thirty-five divisors of flux at most three on the four corners, with the count of states of Lemma 7.5(b) taken as given: the product of the corner signs is $(-1)^d$; the patterns the drive leaves in place; flux one fails clause (iii); flux two cannot meet both clauses; flux three meets clause (iii) always, has one odd corner, and meets clause (ii) exactly when that corner is not P_1; the least flux meeting both is three; the case $D = 3P_4$.

`M1_L7_onedrive.lean` (Proposition 4.5): an integer matrix carrying A_t to its inverse has determinant $-(p^2 + tpq + q^2)$; for $t \geq 4$ the equation $p^2 + tpq + q^2 = -1$ has no solution in integers, by the descent of the proof, so that no such matrix has determinant one; the solution $(1, -1)$ at $t = 3$; the matrix of determinant -1 that carries A_t to its inverse for every t.

The six files follow verbatim. The certificate of the earlier version on the aspect ratio is not part of this paper.

`M1_L1_drive.lean` — the drive

```
/-!
# M1 — Certificate L1: the drive (Section 4)

Paper M1, "Six axioms and their minimal geometry".  Core toolchain,
Lean 4.15.0, no Mathlib, no author axioms, zero `sorry`.

External inputs named here and never used as Lean axioms:—
KatokHasselblatt (hyperbolicity ⇔ |tr| > 2 for SL(2,Z));
Gauss / ——LatimerMacDuffeeTaussky (SL(2,Z) classes of trace t ↔ narrow
classes of binary quadratic forms of discriminant t² − 4).

What is certified: the arithmetic skeleton those theorems consume.
-/

/-- Integer 2×2 matrices. -/
structure M2 where
  a : Int
  b : Int
  c : Int
  d : Int
deriving DecidableEq, Repr

namespace M2
def mul (X Y : M2) : M2 :=
  ⟨X.a*Y.a + X.b*Y.c, X.a*Y.b + X.b*Y.d, X.c*Y.a + X.d*Y.c, X.c*Y.b + X.d*Y.⟩d
def det (X : M2) : Int := X.a*X.d - X.b*X.c
def tr  (X : M2) : Int := X.a + X.d
def I : M2 := ⟨1,0,0,1
```

```
/-- The cat map ₀M = ((2,1),(1,1)). -/
def M0 : M2 := ⟨⟩2,1,1,1
/-- Its inverse −((1,1)−,(1,2)). -/
def M0inv : M2 := ⟨⟩1,-1,-1,2
/-- The quarter-turn S = ((0,1)−,(1,0)), ⁻S¹ = −((0,1),(1,0)). -/
def S : M2 := ⟨⟩0,1,-1,0
def Sinv : M2 := ⟨⟩0,-1,1,0
end M2

open M2

/-- ₀M ∈ SL(2,Z) with trace 3. -/
theorem L1_M0_in_SL2 : det M0 = 1 ∧ tr M0 = 3 := by decide

/-- Smallest case by hand: S ₀M ⁻S¹ = ₀⁻M¹, so the two 3-cycle orientations
    are one conjugacy class. -/
theorem L1_conj_M0_inv : mul (mul S M0) Sinv = M0inv := by decide
theorem L1_M0_M0inv : mul M0 M0inv = I := by decide

/-- Trace 2 is not hyperbolic; the least hyperbolic positive trace is 3:
    t·t > 4 with t > 0 forces t ≥ 3. -/
theorem L1_trace3_min (t : Int) (h : t * t > 4) (hp : t > 0) : t ≥ 3 := by
  rcases (show t = 1 ∨ t = 2 ∨ t ≥ 3 by omega) with h1 | h1 | h1
  · subst h1; omega
  · subst h1; omega
  · exact h1

/-- Discriminant of the trace-3 characteristic polynomial. -/
theorem L1_disc5 : 3 * 3 - 4 = 5 := by decide

/-- Helper: squares grow. -/
theorem sq_ge_of_ge (x k : Int) (hk : 0 ≤ k) (h : k ≤ x) : k*k ≤ x*x :=
  Int.mul_le_mul h h hk (by omega)

/-- m ≥ 0 and (−m1)² < 5 force m ≤ 3. -/
theorem m_le_three (m : Int) (_hm : 0 ≤ m) (h : (m-1)*(m-1) < 5) : m ≤ 3 := by
  rcases (show m ≤ 3 ∨ m ≥ 4 by omega) with h1 | h1
  · exact h1
  · exfalso; have := sq_ge_of_ge (m-1) 3 (by omega) (by omega); omega

/-- (m+1)² > 5 with m ≥ 0 forces m ≥ 2. -/
theorem m_ge_two (m : Int) (hm : 0 ≤ m) (h : (m+1)*(m+1) > 5) : m ≥ 2 := by
  rcases (show m = 0 ∨ m = 1 ∨ m ≥ 2 by omega) with h1 | h1 | h1
  · subst h1; omega
  · subst h1; omega
  · exact h1

/-- Reduced forms of discriminant 5 (Gauss reduction, integer form).  Write
    m = 2·|a|.  (a,b,c) with b² − 4ac = 5, 0 < b, b² < 5, (m+b)² > 5 and
    (−mb)² < 5 forces (a,b,c) ∈ −{(1,1,1), −(1,1,1)}: one reduced cycle,
    hence one class.  This is the hand content of the uniqueness theorem. -/
theorem L1_reduced_disc5 (a b c : Int)
    (hD : b*b - 4*(a*c) = 5) (hb0 : 0 < b) (hb5 : b*b < 5)
    (hlo : (2*(a.natAbs:Int) + b)*(2*(a.natAbs:Int) + b) > 5)
    (hhi : (2*(a.natAbs:Int) - b)*(2*(a.natAbs:Int) - b) < 5) :
    (a = 1 ∧ b = 1 ∧ c = -1) ∨ (a = -1 ∧ b = 1 ∧ c = 1) := by
  have hb : b = 1 ∨ b = 2 := by
    rcases (show b = 1 ∨ b = 2 ∨ b ≥ 3 by omega) with h | h | h
    · exact Or.inl h
    · exact Or.inr h
    · exfalso; have := sq_ge_of_ge b 3 (by omega) h; omega
  rcases hb with hb | hb
  · subst hb
```

```
    have h3 := m_le_three (2*(a.natAbs:Int)) (by omega) hhi
    have h2 := m_ge_two (2*(a.natAbs:Int)) (by omega) hlo
    have hA : (a.natAbs:Int) = 1 := by omega
    have hac : a*c = -1 := by omega
    rcases Int.natAbs_eq a with ha | ha
    · rw [hA] at ha; subst ha; left; refine ⟨rfl, rfl, ?⟩_; omega
    · rw [hA] at ha; subst ha; right; refine ⟨rfl, rfl, ?⟩_; omega
  · subst hb; exfalso; omega

/-- The two reduced forms are one Gauss cycle: ρ−(1,1,1) = −(1,1,1) and
    ρ−(1,1,1) = −(1,1,1), where ρ(a,b,c) = (c, b', (b'−²5)/(4c)) with
    b' ≡ −b (mod 2c) chosen reduced.  Checked by evaluation. -/
theorem L1_cycle_disc5 :
    ((1:Int)*(1:Int) - 5) / (4*(-1)) = (1:Int) ∧ ((1:Int)*(1:Int) - 5) / (4*1) = (-1:Int) := by decide

/-! ### Odd trace ⇔ order three mod 2 (the corner 3-cycle) -/

/-- 2×2 matrices over ₂F = Fin 2. -/
structure F2M where
  a : Fin 2
  b : Fin 2
  c : Fin 2
  d : Fin 2
deriving DecidableEq, Repr

namespace F2M
def mul (X Y : F2M) : F2M :=
  ⟨X.a*Y.a + X.b*Y.c, X.a*Y.b + X.b*Y.d, X.c*Y.a + X.d*Y.c, X.c*Y.b + X.d*Y.⟩d
def I : F2M := ⟨⟩1,0,0,1
def det (X : F2M) : Fin 2 := X.a*X.d - X.b*X.c
def tr  (X : F2M) : Fin 2 := X.a + X.d
def all : List F2M := Id.run do
  let mut l : List F2M := []
  for a in [0,1] do
    for b in [0,1] do
      for c in [0,1] do
        for d in [0,1] do
          l := l ++ ⟨[a,b,c,⟩d]
  return l
end F2M

open F2M in
/-- For every M ∈ SL(2,₂F): M³ = I with M ≠ I  ⇔  tr M = 1 (odd trace).
    Hence the drive 3-cycles the three non-trivial corners iff its trace is odd;
    trace 3 qualifies, every even trace does not.  Finite check over GL(2,₂F). -/
theorem L1_odd_trace_iff_order_three :
    ∀ X ∈ F2M.all, F2M.det X = 1 →
      ((F2M.mul (F2M.mul X X) X = F2M.I ∧ X ≠ F2M.I) ↔ F2M.tr X = 1) := by decide

/-- The cat map reduced mod 2 is ((0,1),(1,1)), of order 3. -/
theorem L1_M0_mod2_order3 :
    F2M.mul (F2M.mul ⟨⟩0,1,1,1 ⟨⟩0,1,1,1) ⟨⟩0,1,1,1 = F2M.I ∧ ⟨⟩(0,1,1,1 : F2M) ≠ F2M.I := by decide

/-! ### The drive is the square of the Fibonacci matrix -/

/-- The Fibonacci matrix F = ((1,1),(1,0)): F·(F_n, F_{−n1}) = (F_{n+1}, F_n). -/
def Fib : M2 := ⟨⟩1,1,1,0

/-- ₀M = F², so the least drive is two Fibonacci steps, and its eigenvalue
    φ² is the square of the Fibonacci growth rate. -/
theorem L1_M0_is_Fib_squared : M2.mul Fib Fib = M0 := by decide

/-- Powers of F carry the Fibonacci numbers: F^n = ((F_{n+1}, F_n), (F_n, F_{−n1})),
```

```
    checked for n = 2..6 (F² = ₀M, F³, ⁴F, ⁵F, ⁶F). -/
theorem L1_Fib_powers :
    M2.mul Fib Fib = ⟨⟩2,1,1,1 ∧
    M2.mul (M2.mul Fib Fib) Fib = ⟨⟩3,2,2,1 ∧
    M2.mul (M2.mul (M2.mul Fib Fib) Fib) Fib = ⟨⟩5,3,3,2 ∧
    M2.mul (M2.mul (M2.mul (M2.mul Fib Fib) Fib) Fib) Fib = ⟨⟩8,5,5,3 ∧
    M2.mul (M2.mul (M2.mul (M2.mul (M2.mul Fib Fib) Fib) Fib) Fib) Fib = ⟨⟩13,8,8,5 := by decide

/-- Fibonacci recursion in the entries: the (1,1) entry of ⁿ⁺¹F is the sum of the
    (1,1) and (1,2) entries of ⁿF. -/
theorem L1_Fib_recursion (X : M2) : (M2.mul X Fib).a = X.a + X.b := by
  simp only [M2.mul, Fib, Int.mul_one, Int.mul_zero, Int.add_zero]

/-- tr F = 1, det F = −1: F itself is not in SL(2,Z); its square is the least
    element of SL(2,Z) with trace > 2. -/
theorem L1_Fib_not_SL2 : M2.det Fib = -1 ∧ M2.tr Fib = 1 ∧ M2.det (M2.mul Fib Fib) = 1 ∧ M2.tr (M2.mul
     Fib Fib) = 3 := by decide

/-- The drive returns exactly one point of the torus to itself: det(₀M − I) = −1,
so ₀M − I is invertible over ℤ and ₀Mx ≡ x (mod ℤ²) forces x ≡ 0. Added for
M1 v3 (the place we see from, Definition "shadow, formal"). -/
theorem L1_one_fixed_point : M2.det ⟨2 - 1, 1, 1, 1 - ⟩1 = -1 := by decide
```

M1_L2_fibre.lean — the fibre

```
/-!
# M1 — Certificate L2: the fibre algebra and the four corners (Section 5)

Core toolchain, Lean 4.15.0, no Mathlib, no author axioms, zero `sorry`.
External inputs named, never Lean axioms: Frobenius 1878 (finite-dimensional
associative real division algebras are ℝ, ℂ, ℍ); Thurston (Euclidean
2-orbifolds).

Certified: the Hamilton relations in an integer quaternion model; the fixed
subalgebras of conjugation and of the inner automorphisms Ad_i, Ad_j, Ad_k;
the order-three automorphism cycling i → j → k; the corner count 4 = 2^2;
the orbifold Euler characteristic of the pillowcase.
-/

/-- Quaternions with integer coefficients a + b i + c j + d k. -/
structure Q where
  a : Int
  b : Int
  c : Int
  d : Int
deriving DecidableEq, Repr

namespace Q
def mul (p q : Q) : Q :=
  ⟨p.a*q.a - p.b*q.b - p.c*q.c - p.d*q.d,
   p.a*q.b + p.b*q.a + p.c*q.d - p.d*q.c,
   p.a*q.c - p.b*q.d + p.c*q.a + p.d*q.b,
   p.a*q.d + p.b*q.c - p.c*q.b + p.d*q.⟩a
def conj (q : Q) : Q := ⟨q.a, -q.b, -q.c, -q.⟩d
def one : Q := ⟨⟩1,0,0,0
def i : Q := ⟨⟩0,1,0,0
def j : Q := ⟨⟩0,0,1,0
def k : Q := ⟨⟩0,0,0,1
def neg (q : Q) : Q := ⟨-q.a,-q.b,-q.c,-q.⟩d
/-- Norm form N(q) = q q̄ (real part). -/
def norm (q : Q) : Int := q.a*q.a + q.b*q.b + q.c*q.c + q.d*q.d
```

```
/-- Inner automorphism by a unit u with u² = −1: Ad_u(q) = u q ⁻u¹ = −u q u. -/
def Ad (u q : Q) : Q := neg (mul (mul u q) u)
end Q

open Q

/-- Hamilton's relations. -/
theorem L2_hamilton :
    mul i i = neg one ∧ mul j j = neg one ∧ mul k k = neg one ∧
    mul i j = k ∧ mul j k = i ∧ mul k i = j ∧
    mul j i = neg k ∧ mul k j = neg i ∧ mul i k = neg j := by decide

/-- Non-commutativity witnessed. -/
theorem L2_noncomm : mul i j ≠ mul j i := by decide

/-- The norm is q q̄ and is positive-definite: N(q) = 0 ⇒ q = 0.
    (This is the step where A1 — compact corner unit groups, u² = −1 — makes
    the algebra a division algebra; cf. Section 5.3.) -/
theorem L2_norm_is_qqbar (q : Q) : (mul q (conj q)).a = norm q := by
  cases q with | mk a b c d =>
  show a*a - b*(-b) - c*(-c) - d*(-d) = a*a + b*b + c*c + d*d
  rw [Int.mul_neg, Int.mul_neg, Int.mul_neg]
  omega

/-- x·x ≥ 0 over the integers (core proof). -/
theorem sq_nonneg (x : Int) : 0 ≤ x*x := by
  rcases Int.le_total 0 x with h | h
  · exact Int.mul_nonneg h h
  · have := Int.mul_nonneg_of_nonpos_of_nonpos h h; exact this

/-- The norm is positive-definite: N(q) = 0 ⇒ q = 0.  (Section 5.3: this is
    where A1 — compact corner unit groups, u² = −1 — makes ℍ a division algebra.) -/
theorem L2_norm_posdef (q : Q) (h : norm q = 0) : q = ⟨0,0,0,0⟩ := by
  have ha := sq_nonneg q.a; have hb := sq_nonneg q.b
  have hc := sq_nonneg q.c; have hd := sq_nonneg q.d
  simp [norm] at h
  have a0 : q.a*q.a = 0 := by omega
  have b0 : q.b*q.b = 0 := by omega
  have c0 : q.c*q.c = 0 := by omega
  have d0 : q.d*q.d = 0 := by omega
  have key : ∀ x : Int, x*x = 0 → x = 0 := by
    intro x hx
    rcases Int.mul_eq_zero.mp hx with h | h <;> exact h
  cases q
  simp only at a0 b0 c0 d0
  exact (by simp [key _ a0, key _ b0, key _ c0, key _ d0])

/-- Fixed set of conjugation is ℝ: q̄ = q ⇔ b = c = d = 0. -/
theorem L2_fix_conj (q : Q) : conj q = q ↔ (q.b = 0 ∧ q.c = 0 ∧ q.d = 0) := by
  cases q; simp [conj]; omega

/-- Fixed set of Ad_i is ℂ_i: Ad_i(q) = q ⇔ c = d = 0. -/
theorem L2_fix_Ad_i (q : Q) : Ad i q = q ↔ (q.c = 0 ∧ q.d = 0) := by
  cases q; simp [Ad, mul, neg, i]; omega
theorem L2_fix_Ad_j (q : Q) : Ad j q = q ↔ (q.b = 0 ∧ q.d = 0) := by
  cases q; simp [Ad, mul, neg, j]; omega
theorem L2_fix_Ad_k (q : Q) : Ad k q = q ↔ (q.b = 0 ∧ q.c = 0) := by
  cases q; simp [Ad, mul, neg, k]; omega

/-- Conjugation composed with Ad_u fixes the 3-dimensional u⊥^, which is
    not closed under multiplication (j·k = i ∉ i⊥^): so ℝ{, ℂ_i, ℂ_j, ℂ_k}
    are the only commutative fixed subalgebras. -/
theorem L2_fix_conj_Ad_i (q : Q) : conj (Ad i q) = q ↔ q.b = 0 := by
```

```
  cases q; simp [Ad, mul, neg, conj, i]; omega
theorem L2_iperp_not_closed : (mul j k).b ≠ 0 := by decide

/-- The order-three automorphism cycling i → j → k → i.  With 2w = 1+i+j+k
    and (2w)⁻¹ = −(1−−ijk)/4:  (2w)·i·(2w)‾= 4j, etc. -/
def twow : Q := ⟨⟩1,1,1,1
theorem L2_Ad_w_cycles :
    mul (mul twow i) (conj twow) = ⟨⟩0,0,4,0 ∧
    mul (mul twow j) (conj twow) = ⟨⟩0,0,0,4 ∧
    mul (mul twow k) (conj twow) = ⟨⟩0,4,0,0 := by decide
/-- w³ = −1 (as (2w)³ = −8), so Ad_w has order 3. -/
theorem L2_w_cubed : mul (mul twow twow) twow = ⟨⟩-8,0,0,0 := by decide

/-- The corner group: the eight units {±1, ±i, ±j, ±k} modulo sign give
    exactly four subalgebras.  Represented as the four generators 1, i, j, k
    each fixed (up to sign) by negation. -/
theorem L2_corner_count : [one, i, j, k].length = 4 ∧ (8:Nat) / 2 = 4 := by decide

/-- Two-torsion of a torus ⁿT has ⁿ2 points; four corners force n = 2. -/
theorem L2_two_torsion (n : Nat) (h : 2 ^ n = 4) : n = 2 := by
  match n with
  | 0 => simp at h
  | 1 => simp at h
  | 2 => rfl
  | 3 => simp at h
  | n+4 =>
    have : 2 ^ (n+4) = 16 * 2 ^ n := by rw [Nat.pow_add]; omega
    have hp : 2 ^ n ≥ 1 := Nat.one_le_two_pow
    omega

/-- Orbifold Euler characteristic of a sphere with four cone points of order 2,
    in integer form (doubled): 2·2 − 4·1 = 0; hence Euclidean (Thurston). -/
theorem L2_chi_orb_zero : 2*2 - 4*1 = 0 := by decide

/-- χ(T²ℤ₂/) = ½χ((T²) + #fixed) = ½(0 + 4) = 2, and by orbifold
    −GaussBonnet πχ2 = 4 · (deficit π) ⇒ χ = 2. -/
theorem L2_chi_pillowcase : (0 + 4) / 2 = 2 ∧ 4 / 2 = 2 := by decide
```

M1_L3_dimension.lean — the dimension

```
/-!
# M1 − Certificate L3: dimension twelve and the split (Section 6)
Core toolchain, Lean 4.15.0, no Mathlib, no author axioms, zero `sorry`.
External inputs named, never Lean axioms: Salamon 1982 (almost-quaternionic
structures live in dimension 4n); the identity Sp(1)Sp(1) = SO(4).
-/

/-- The −timespace block T × Σ is the quaternions: dimension 4.
    With D = 4 + dim F and F of quaternionic dimension m ≥ 2 (A3 non-vacuous
    on F), D ≥ 12; every D ≡ 0 (mod 4) below 12 fails. -/
theorem L3_dim_ge_twelve (D m : Nat) (h4 : 4 ∣ D) (hF : D = 4 + 4*m) (hm : m ≥ 2) : D ≥ 12 := by
  omega

/-- Dimensions 4 ≤ D ≤ 11 all fail: either 4 ∤ D, or D ∈ {4, 8} where
    F has quaternionic dimension 0 or 1. -/
theorem L3_below_twelve (D : Nat) (hlo : 4 ≤ D) (hhi : D ≤ 11) :
    ¬ (4 ∣ D) ∨ D = 4 ∨ D = 8 := by
  rcases (show D = 4 ∨ D = 5 ∨ D = 6 ∨ D = 7 ∨ D = 8 ∨ D = 9 ∨ D = 10 ∨ D = 11 by omega)
    with h | h | h | h | h | h | h | h <;> subst h <;> decide

/-- Dimension 12 is admissible: 4 ∣ 12 and F has quaternionic dimension 2. -/
```

```
theorem L3_twelve_ok : 4 ∣ 12 ∧ 12 = 4 + 4*2 := by decide

/-- Within D = 12 the split dim Σ + dim B = 9 with T × Σ quaternionic
    (dim Σ + 1 ≡ 0 mod 4), B × T²ℤ₂/ quaternionic (dim B + 2 ≡ 0 mod 4), and
    F = B × T²ℤ₂/ of quaternionic dimension ≥ 2 (dim B + 2 ≥ 8, the same
    condition that excludes D = 8), has the unique solution (3, 6).
    Without the last hypothesis (7, 2) also satisfies the congruences — a
    fact this certificate surfaced; it is excluded by A3 on F. -/
theorem L3_split_three_six (s b : Nat)
    (hsum : s + b = 9) (hs : (s + 1) % 4 = 0) (hb : (b + 2) % 4 = 0)
    (hs1 : s ≥ 1) (hF : b + 2 ≥ 8) : s = 3 ∧ b = 6 := by
  have hs' : s = 3 ∨ s = 7 := by omega
  rcases hs' with h | h
  · subst h; exact ⟨rfl, by ⟩omega
  · subst h; exfalso; omega

/-- The excluded alternative, recorded: (7, 2) meets the congruences and sum. -/
theorem L3_seven_two_meets_congruences : (7 + 2 = 9) ∧ ((7 + 1) % 4 = 0) ∧ ((2 + 2) % 4 = 0) ∧ ¬ (2 + 2
     ≥ 8) := by decide

/-- Euler characteristic of the internal factor: χℂ(P²)χ·(S²)χ·(T²ℤ₂/) = 3·2·2 = 12,
    and the dimension identity 1 + 3 + 8 = 12. -/
theorem L3_chi_K8 : 3 * 2 * 2 = 12 ∧ 1 + 3 + 8 = 12 := by decide
```

M1_L4_base.lean — the base

```
/-!
# M1 — Certificate L4: the base, the Spin^c twist, the determinant line (Section 7)
Core toolchain, Lean 4.15.0, no Mathlib, no author axioms, zero `sorry`.
External inputs named, never Lean axioms: —LawsonMichelsohn (Spin^c criterion₁
c(L) ≡ ₂w mod 2); the Euler sequence on ℂP².
-/

/-- The sphere of imaginary units: a + b i + c j + d k with a = 0 and
    b² + c² + d² = 1 has exactly the six integer points ±i, ±j, ±k, i.e. the
    three coordinate axes — the three units — on S². -/
theorem L4_unit_sphere_axes (b c d : Int) (h : b*b + c*c + d*d = 1) :
    (b = 0 ∧ c = 0 ∧ (d = 1 ∨ d = -1)) ∨ (b = 0 ∧ d = 0 ∧ (c = 1 ∨ c = -1)) ∨
    (c = 0 ∧ d = 0 ∧ (b = 1 ∨ b = -1)) := by
  have sq : ∀ x : Int, 0 ≤ x*x := by
    intro x; rcases Int.le_total 0 x with hx | hx
    · exact Int.mul_nonneg hx hx
    · exact Int.mul_nonneg_of_nonpos_of_nonpos hx hx
  have big : ∀ x : Int, (x ≥ 2 ∨ x ≤ -2) → x*x ≥ 4 := by
    intro x hx; rcases hx with hx | hx
    · exact Int.mul_le_mul hx hx (by omega) (by omega)
    · have h1 : (-x) ≥ 2 := by omega
      have := Int.mul_le_mul h1 h1 (by omega) (by omega)
      have e : (-x)*(-x) = x*x := by simp [Int.neg_mul_neg]
      omega
  have hb := sq b; have hc := sq c; have hd := sq d
  have bb : b = -1 ∨ b = 0 ∨ b = 1 := by
    rcases (show (b ≥ 2 ∨ b ≤ -2) ∨ (b = -1 ∨ b = 0 ∨ b = 1) by omega) with h1 | h1
    · have := big b h1; omega
    · exact h1
  have cc : c = -1 ∨ c = 0 ∨ c = 1 := by
    rcases (show (c ≥ 2 ∨ c ≤ -2) ∨ (c = -1 ∨ c = 0 ∨ c = 1) by omega) with h1 | h1
    · have := big c h1; omega
    · exact h1
  have dd : d = -1 ∨ d = 0 ∨ d = 1 := by
    rcases (show (d ≥ 2 ∨ d ≤ -2) ∨ (d = -1 ∨ d = 0 ∨ d = 1) by omega) with h1 | h1
```

```
    · have := big d h1; omega
    · exact h1
  rcases bb with hb' | hb' | hb' <;> rcases cc with hc' | hc' | hc' <;> rcases dd with hd' | hd' | hd'
    <;>
    subst hb' <;> subst hc' <;> subst hd' <;> simp at h ⊢

/-- Total Chern class of ℂP²: (1+h)³ = 1 + 3h + 3h² + h³, h³ = 0;
    (1+h)³ expanded as a polynomial in h with h³ = 0: coefficients (1,3,3). -/
def cube_coeffs : List Int := [1, 3, 3]   -- 1 + 3h + 3h²  (h³ dropped)
theorem L4_chern_CP2 : cube_coeffs = [1, 3, 3] ∧ (1+1)*(1+1)*(1+1) = (8:Int) ∧ 1 + 3 + 3 + 1 = 8 := by
    decide

/-- ₂w₂(P²) = ₁c mod 2 = 3 mod 2 = 1 ≠ 0: not spin.  The congruence
    ₁c(L) ≡ ₂w holds for L = 𝒪(3): 3 ≡ 1 (mod 2). -/
theorem L4_not_spin_and_congruence : (3:Nat) % 2 = 1 ∧ (3:Nat) % 2 = (3:Nat) % 2 ∧ (0:Nat) % 2 = 0 := by
    decide

/-- Odd lines 𝒪(1), 𝒪(3), 𝒪(5) all satisfy the congruence; the canonical
    (untwisted) one is ⁻K¹ = 𝒪(3), the others are 𝒪(3) ⊗ 𝒪(e)² with e ≠ 0:
    3 + 2e for e = −1, 1 gives 1, 5. -/
theorem L4_odd_lines : 3 + 2*(-1:Int) = 1 ∧ 3 + 2*(1:Int) = 5 ∧ ∀( e : Int, (3 + 2*e) % 2 = 1) := by
  refine ⟨by decide, by decide, ?⟩_; intro e; omega

/-- Flux parity: Q_c odd on ℂP², Q_w even on S² (instances of ₁c ≡ ₂w). -/
theorem L4_flux_parity : (3:Nat) % 2 = 1 ∧ (6:Nat) % 2 = 0 := by decide

/-- Four-corner product of the Spin^c involution lift −(1)^{mn} over
    m, n ∈ {0,1}: three corners +1, one corner −1, product −1. -/
def cornerPhase (m n : Nat) : Int := if m*n % 2 = 0 then 1 else -1
theorem L4_four_corner_product :
    cornerPhase 0 0 * cornerPhase 1 0 * cornerPhase 0 1 * cornerPhase 1 1 = -1 ∧
    cornerPhase 0 0 = 1 ∧ cornerPhase 1 0 = 1 ∧ cornerPhase 0 1 = 1 ∧ cornerPhase 1 1 = -1 := by decide

/-- Euler characteristic of the base factors and of ℂP² as a count of units:
    χ(ℂP²) = 3 = number of imaginary units; χ(S²) = 2. -/
theorem L4_chi_counts : (3:Nat) = [1,1,1].length ∧ (2:Nat) = [1,1].length := by decide
```

M1_L6_twist.lean — the twist of the fibre

```
/-!
# M1 — Certificate L6: the twist of the fibre (Section 7.4)

Paper M1, "Six axioms and their minimal geometry".  Core Lean, no Mathlib,
no author axioms, zero `sorry`.

External input named here and never used as a Lean axiom: the classical
theory of elliptic functions (an even elliptic function is a rational
function of the Weierstrass function), which gives the count of states
`1 + Σ ⌊₁a⌋/2` kept by the first lift of the fold (Lemma 7.5 (b)).

What is certified: the finite case analysis of Proposition 7.6, taking that
count as given.  A divisor on the four corners is a quadruple of
multiplicities; the drive fixes ₁P and cycles ₂P → ₃P → ₄P.

What is not certified: the count itself, and the reading that the states of
the fibre are the objects of the fifth axiom.
-/

/-- A divisor on the four corners: multiplicities at ₁P, ₂P, ₃P, ₄P. -/
structure CDiv where
  a1 : Nat
```

```
  a2 : Nat
  a3 : Nat
  a4 : Nat
deriving DecidableEq, Repr

namespace CDiv
/-- The flux. -/
def deg (D : CDiv) : Nat := D.a1 + D.a2 + D.a3 + D.a4
/-- States kept by the first lift of the fold: 1 + Σ ⌊ᵢa⌋/2. -/
def kept1 (D : CDiv) : Nat := 1 + D.a1 / 2 + D.a2 / 2 + D.a3 / 2 + D.a4 / 2
/-- States kept by the second lift: the remainder. -/
def kept2 (D : CDiv) : Nat := D.deg - D.kept1
/-- Corner signs of the first lift; `true` stands for −1. -/
def sgn (D : CDiv) : List Bool := [D.a1 % 2 == 1, D.a2 % 2 == 1, D.a3 % 2 == 1, D.a4 % 2 == 1]
/-- Corner signs up to one overall sign: the sign at ₁P is made +1. -/
def pat (D : CDiv) : List Bool := if D.a1 % 2 == 1 then D.sgn.map (fun b => !b) else D.sgn
/-- One step of the drive: ₂P → ₃P → ₄P → ₂P, ₁P fixed. -/
def step (D : CDiv) : CDiv := ⟨D.a1, D.a4, D.a2, D.⟩a3
/-- The corner signs are carried to themselves by one step. -/
def fixedByDrive (D : CDiv) : Bool := D.step.pat == D.pat
/-- Clause (ii) of the fifth axiom: the structure is not carried to itself. -/
def clauseII (D : CDiv) : Bool := !D.fixedByDrive
/-- Clause (iii): two states on the pillowcase, for one of the two lifts. -/
def clauseIII (D : CDiv) : Bool := decide (2 ≤ D.kept1) || decide (2 ≤ D.kept2)
/-- Number of corners with sign −1. -/
def minus (D : CDiv) : Nat := (D.sgn.filter (fun b => b)).length
/-- All divisors of flux at most three (thirty-five of them). -/
def all : List CDiv :=
  (List.range 4).flatMap fun a => (List.range (4 - a)).flatMap fun b =>
  (List.range (4 - a - b)).flatMap fun c => (List.range (4 - a - b - c)).map fun d => ⟨a, b, c, ⟩d
end CDiv

open CDiv

set_option maxRecDepth 8192

/-- The list has thirty-five members. -/
theorem L6_count : CDiv.all.length = 35 := by decide

/-- The product of the four corner signs is −(1)^flux. -/
theorem L6_corner_product : ∀ D ∈ CDiv.all, D.minus % 2 = D.deg % 2 := by decide

/-- Three steps of the drive return every pattern. -/
theorem L6_return_three : ∀ D ∈ CDiv.all, D.step.step.step = D := by decide

/-- A pattern is left in place by the drive exactly when all corners are
    alike or the odd corner is ₁P. -/
theorem L6_fixed_patterns : ∀ D ∈ CDiv.all,
    (D.fixedByDrive = true ↔
      (D.pat = [false, false, false, false] ∨ D.pat = [false, true, true, true])) := by decide

/-- Flux one: clause (iii) fails. -/
theorem L6_flux_one : ∀ D ∈ CDiv.all, D.deg = 1 → D.clauseIII = false := by decide

/-- Flux two: clause (iii) forces a structure the drive leaves in place,
    so clause (ii) fails. -/
theorem L6_flux_two : ∀ D ∈ CDiv.all, D.deg = 2 → D.clauseIII = true → D.clauseII = false := by decide

/-- Flux three: clause (iii) always holds, one lift keeping two states and
    removing one. -/
theorem L6_flux_three_states : ∀ D ∈ CDiv.all, D.deg = 3 →
    (D.clauseIII = true ∧ ((D.kept1 = 2 ∧ D.kept2 = 1) ∨ (D.kept1 = 1 ∧ D.kept2 = 2))) := by decide
```

```
/-- Flux three: exactly one corner is odd (up to the overall sign). -/
theorem L6_flux_three_one_odd : ∀ D ∈ CDiv.all, D.deg = 3 →
    (D.pat = [false, false, false, true] ∨ D.pat = [false, false, true, false] ∨
     D.pat = [false, true, false, false] ∨ D.pat = [false, true, true, true]) := by decide

/-- Flux three: clause (ii) holds exactly when the odd corner is not ₁P. -/
theorem L6_flux_three_cycled : ∀ D ∈ CDiv.all, D.deg = 3 →
    (D.clauseII = true ↔ D.pat ≠ [false, true, true, true]) := by decide

/-- No flux of one or two meets both clauses; flux three does. -/
theorem L6_least_flux :
    ∀( D ∈ CDiv.all, D.deg = 1 ∨ D.deg = 2 → ¬ (D.clauseII = true ∧ D.clauseIII = true)) ∧
    ∃( D ∈ CDiv.all, D.deg = 3 ∧ D.clauseII = true ∧ D.clauseIII = true) := by decide

/-- The smallest case: D = 3 ₄P has two states kept, one removed, corner
    signs (+, +, +, −), and meets both clauses. -/
theorem L6_smallest_case :
    ⟨(0, 0, 0, ⟩3 : CDiv).kept1 = 2 ∧ ⟨(0, 0, 0, ⟩3 : CDiv).kept2 = 1 ∧
    ⟨(0, 0, 0, ⟩3 : CDiv).sgn = [false, false, false, true] ∧
    ⟨(0, 0, 0, ⟩3 : CDiv).clauseII = true ∧ ⟨(0, 0, 0, ⟩3 : CDiv).clauseIII = true := by decide
```

M1_L7_onedrive.lean — one drive at trace three only

```
/-!
# M1 — Certificate L7: one drive at trace three only (Proposition 4.5)

Paper M1, "Six axioms and their minimal geometry".  Core library only,
no Mathlib, no author axioms, zero `sorry`.  Written for Lean 4.29.1; it uses
the core tactics `omega`, `grind` and `push_cast`.

What is certified.  With A_t = ((t,-1),(1,0)) and ⁻A_t¹ = ((0,1),(-1,t)):
* an integer matrix P = ((p,q),(r,s)) with P A_t = ⁻A_t¹ P has r = t p + q,
  s = -p and det P = -(p² + t p q + q²);
* for t ≥ 4 the equation p² + t p q + q² = -1 has no solution in integers
  (the descent of the proof of Proposition 4.5), so that no such P has
  determinant 1;
* at t = 3 it has the solution (1,-1);
* P = ((1,0),(t,-1)) has determinant -1 and satisfies P A_t = ⁻A_t¹ P for
  every t: the hypothesis on orientation is used.

What is not certified: that conjugacy classes of SL(2,Z) are the drives up to
a relabelling of the torus that keeps its orientation, and that clause (a) of
Definition 3.2 is applied to them.  Those are the text's.
-/

/-- The descent.  For t ≥ 4 there are no natural numbers x ≥ y with
x² + y² + 1 = t x y.  Induction on a bound for x + y. -/
theorem L7_descent (t : Nat) (ht : 4 ≤ t) :
    ∀ n x y : Nat, x + y ≤ n → y ≤ x → x*x + y*y + 1 = t*x*y → False := by
  intro n
  induction n with
  | zero =>
    intro x y hn hyx h
    have hx : x = 0 := by omega
    have hy : y = 0 := by omega
    subst hx; subst hy
    simp at h
  | succ n ih =>
    intro x y hn hyx h
    -- y is positive
    have hy : 1 ≤ y := by
```

```
      rcases Nat.eq_zero_or_pos y with h0 | h0
      · subst h0; simp at h
      · exact h0
    have hx : 1 ≤ x := by omega
    rcases Nat.lt_or_eq_of_le hyx with hlt | heq
    · -- x > y: pass to the second root x' = t y - x
      have hxx : x * x < x * (t * y) := by
        have : x * (t * y) = t * x * y := by grind
        omega
      have hxty : x < t * y := Nat.lt_of_mul_lt_mul_left hxx
      -- write t y = x + x'
      obtain ⟨x', hx⟩' : ∃ x', t * y = x + x' := ⟨t * y - x, by ⟩omega
      have hprod : x * x' = y * y + 1 := by
        have : t * x * y = x * (x + x') := by rw ←[ hx']; grind
        grind
      have hx'y : x' ≤ y := by
        rcases Nat.lt_or_ge y x' with hgt | hle
        · have h1 : (y + 1) * (y + 1) ≤ x * x' :=
            Nat.mul_le_mul (by omega) (by omega)
          have h2 : (y + 1) * (y + 1) = y * y + 2 * y + 1 := by grind
          omega
        · exact hle
      have hnew : y * y + x' * x' + 1 = t * y * x' := by
        have : t * y * x' = (x + x') * x' := by rw ←[ hx']
        grind
      exact ih y x' (by omega) hx'y hnew
    · -- x = y: then x² (t - 2) = 1, impossible for t ≥ 4
      subst heq
      have h4 : 4 * (y * y) ≤ t * (y * y) := Nat.mul_le_mul_right _ ht
      have h5 : t * y * y = t * (y * y) := by grind
      have h6 : 1 ≤ y * y := Nat.mul_le_mul hy hy
      omega

/-- No natural solution, in either order. -/
theorem L7_no_nat_solution (t : Nat) (ht : 4 ≤ t) (x y : Nat) :
    x*x + y*y + 1 ≠ t*x*y := by
  intro h
  rcases Nat.le_total y x with hyx | hxy
  · exact L7_descent t ht (x + y) x y (Nat.le_refl _) hyx h
  · exact L7_descent t ht (y + x) y x (Nat.le_refl _) hxy (by grind)

/-- For t ≥ 4 the equation p² + t p q + q² = -1 has no solution in integers. -/
theorem L7_no_int_solution (t : Nat) (ht : 4 ≤ t) (p q : Int) :
    p*p + (t:Int)*p*q + q*q ≠ -1 := by
  intro h
  obtain ⟨x, ⟩hx := Int.eq_nat_or_neg p
  obtain ⟨y, ⟩hy := Int.eq_nat_or_neg q
  have key : ∀ x y : Nat, (x:Int)*x + (t:Int)*x*y + y*y ≠ -1 := by
    intro x y hxy
    have : (0:Int) ≤ ((x*x + t*x*y + y*y : Nat) : Int) := Int.natCast_nonneg _
    push_cast at this
    omega
  have key2 : ∀ x y : Nat, (x:Int)*x - (t:Int)*x*y + y*y ≠ -1 := by
    intro x y hxy
    apply L7_no_nat_solution t ht x y
    have : ((x*x + y*y + 1 : Nat) : Int) = ((t*x*y : Nat) : Int) := by
      push_cast; omega
    exact_mod_cast this
  rcases hx with hx | hx <;> rcases hy with hy | hy <;> subst hx <;> subst hy
  · exact key x y h
  · exact key2 x y (by grind)
  · exact key2 x y (by grind)
  · exact key x y (by grind)
```

```
/-- Entry comparison: P A_t = ⁻A_t¹ P gives r = t p + q and s = -p, and then
det P = -(p² + t p q + q²). -/
theorem L7_det_of_intertwiner (t p q r s : Int)
    (e11 : p*t + q = r) (e12 : -p = s) :
    p*s - q*r = -(p*p + t*p*q + q*q) := by
  subst e11; subst e12; grind

/-- For t ≥ 4 no integer matrix of determinant 1 conjugates A_t to its inverse. -/
theorem L7_not_conj_inverse (t : Nat) (ht : 4 ≤ t) (p q r s : Int)
    (e11 : p*t + q = r) (e12 : -p = s) : p*s - q*r ≠ 1 := by
  intro hdet
  have h := L7_det_of_intertwiner t p q r s e11 e12
  exact L7_no_int_solution t ht p q (by omega)

/-- A_t ⁻A_t¹ = 1, entry by entry. -/
theorem L7_inverse (t : Int) :
    t*0 + (-1)*(-1) = 1 ∧ t*1 + (-1)*t = 0 ∧ (1:Int)*0 + 0*(-1) = 0 ∧ (1:Int)*1 + 0*t = 1 := by
  refine ⟨by omega, by omega, by omega, by ⟩omega

/-- At t = 3 the equation has the solution (1,-1). -/
theorem L7_trace3_solution : (1:Int)*1 + 3*1*(-1) + (-1)*(-1) = -1 := by decide

/-- P = ((1,0),(t,-1)) has determinant -1 and satisfies P A_t = ⁻A_t¹ P:
the four entries of the two products agree. -/
theorem L7_flip (t : Int) :
    (1:Int)*(-1) - 0*t = -1 ∧
    (1*t + 0*1 = 0*1 + 1*t) ∧ ((1:Int)*(-1) + 0*0 = 0*0 + 1*(-1)) ∧
    (t*t + (-1)*1 = (-1)*1 + t*t) ∧ (t*(-1) + (-1)*0 = (-1)*0 + t*(-1)) := by
  refine ⟨by omega, by omega, by omega, by omega, by ⟩omega
```

B Names and glossary

This appendix fixes the names used in this paper. The four corners are named first. The terms follow: each entry gives the term, the words of this paper that define it, copied as they stand, and the place where they stand. A later paper of the series carries this glossary, keeps its names, and adds the terms it introduces.

The four corners

The four corners are named by position, $P_1 = (0,0)$, $P_2 = (\frac{1}{2},0)$, $P_3 = (0,\frac{1}{2})$, $P_4 = (\frac{1}{2},\frac{1}{2})$ in coordinates $(\theta_\psi, \theta_\omega)$, and by algebra, $C_0 = \mathbb{R}$, C_i, C_j, C_k. Throughout the programme the dictionary is: $P_1 = C_0$ (fixed by the drive; ι acts as conjugation); $P_2 = C_i$ (the ψ half-period); $P_3 = C_j$ (the ω half-period); $P_4 = C_k$ (the corner at which the Spin^c lift carries the sign -1). The drive fixes P_1; the lift distinguishes P_4; these are different structures and distinguish different corners. The drive carries the lift's odd corner $P_4 \to P_2 \to P_3 \to P_4$, so that the sign pattern returns after three steps and the time circle is three steps (Corollary 7.8); the dictionary gives the pattern at the step called the first. The assignment of i and j between P_2 and P_3 is the orientation of the three-cycle, which Theorem 4.2 shows to be a single conjugacy class and which no equivariant invariant can detect: it is a convention and carries no content.

The axioms and the tags

The six axioms of Section 1.2 are cited as A1 to A6: closure (A1), growth and decay (A2), order (A3), drive (A4), shadow (A5), oneness (A6). "Rigour is tagged [R] (proved, or a cited classical theorem), [D] (derived under a stated reading) and [M] (motivated)." (Section 1.4)

Terms

The terms stand in the order in which the paper first defines them. A sentence after the brackets, where there is one, says what a symbol or a word in the quoted words stands for, or where it is defined.

geometry

"A *geometry* here is a smooth shape, of some number of dimensions, on which lengths and angles can be measured, and which is allowed to have a few isolated sharp points (cone points, like the tip of a paper cone)." (Section 2.1)

the pieces T, Σ, F; the geometry M

"T is a one-dimensional piece carrying time, Σ is a piece carrying space, and F is an internal piece" (Definition 2.1) The geometry is their product, $M = (T \times \Sigma) \times F$, equation (2).

shadow; the projection π

"the image under π of a point or of a subset of M being its *shadow*" (Definition 2.1) Here $\pi\colon M \to T \times \Sigma$, $\pi(t, x, f) = (t, x)$, is the projection that forgets F.

warp

"the size of one piece depends on where one is in another, the way the circles of latitude on a globe shrink towards the poles" (Section 2.3)

the fold ι

"there is a symmetry of the shape that swaps κ for $-\kappa$, and applied twice it is an involution ι of the warped piece" … "Section 5 will find that it is the fold that halves the internal piece and, on the algebra of the next axiom, the operation of conjugation" … "Under clause (c) of Definition 3.2 the fibre is the torus T^2, on which ι is inversion $z \mapsto -z$ and the drive is a hyperbolic element of $\mathrm{SL}_2(\mathbb{Z})$." (Definition 2.4, Section 2.3 and Proposition 5.5) Here κ is the constant rate of the growth law of Definition 2.4, and $T^2 = \mathbb{R}^2/\mathbb{Z}^2$ is the fibre of F carrying the drive; the drive and clause (c) are defined below.

involution

"An involution is a symmetry that undoes itself when applied twice, like a reflection." (Section 2.3)

fibre algebra $\mathcal{A}$

"The *fibre algebra* $\mathcal{A}$ of a geometry is the collection of operations on the fibres of F, with the product of two operations being the result of doing one and then the other, in the order in which they occur in time." (Definition 2.5) The internal piece is a base with a fibre over it: in the geometry derived, $F = \text{base} \times T^2/\mathbb{Z}_2$ (Remark 6.3).

quaternions $\mathbb{H}$; the units i, j, k

"the quaternions, the four-dimensional number system with units $1, i, j, k$ and $i^2 = j^2 = k^2 = ijk = -1$, in which $ij = k$ but $ji = -k$" (Section 2.4)

realised

"A homogeneous space G/H formed from the symmetry group G of the fibre algebra and the stabiliser H of a structure selected on it is *realised* by the geometry: it appears as a factor of the base over which F sits." (Definition 2.8) A symmetry is defined in the next entry.

symmetry (of the fibre algebra)

"A symmetry here is a map of the imaginary units that keeps their length and is not asked to keep their products" (Definition 2.8)

drive

"A geometry satisfies A4 if $T \times F$ carries a smooth flow φ_t with no fixed points and positive topological entropy." … "the drive is what time does to the internal piece as it goes round" (Definition 2.9 and Section 2.5)

modulus

"A *modulus* of a geometry satisfying A1–A5 is a continuous parameter of its structure data (the lengths, with the warp of Definition 2.4 and its rate κ, the complex and Spin^c structures, the orbifold data, the flux integers) that the six axioms and the readings of this paper leave free, taken modulo diffeomorphism and overall scale." (Definition 3.1) Flux is defined below, under "twist of the fibre; flux".

minimal

"A geometry satisfying A1–A5 satisfies A6, and is called *minimal*, if it is selected by the following three clauses." (Definition 3.2) The three clauses are (a), (b) and (c) of that definition.

the same (of two geometries); relabelling

"Two geometries satisfying A1–A5 are *the same* when a relabelling carries one to the other: a one-to-one map of the one shape onto the other, smooth in both directions away from the cone points" (Definition 3.3) The conditions the map must meet follow in that definition.

unique

"A geometry of a given kind is *unique* when any two of that kind are the same." (Definition 3.3)

fibre torus T^2; step (of the drive)

"The drive of A4 acts on T^2 by its return map, an element $M \in \mathrm{SL}_2(\mathbb{Z})$ with trace t. One application of this map is a *step* of the drive." (Section 4) Here $T^2 = \mathbb{R}^2/\mathbb{Z}^2$ is the fibre of F carrying the drive. The letter M names this map in Section 4 and the geometry elsewhere; which is meant is clear from its place.

Fibonacci matrix F; M_0

"The Fibonacci matrix $F = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ has determinant -1: it reverses the handedness and is not a step. Its square $M_0 = F^2 = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ has determinant $(-1)^2 = +1$ and is the least power of F that is a step." (Section 4) The handedness is the one that Definition 2.7 fixes on the

internal piece. The letter F also names the internal piece; which is meant is clear from its place.

quarter-turn S

"the quarter-turn $S = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$" (the proof of Theorem 4.2)

anti-involution $a \mapsto a^*$

"Let $\mathcal{A}$ be an associative real algebra with unit $1 \neq 0$, and let ι act on it as an anti-involution $a \mapsto a^*$ (the algebraic image of exchanging growth and decay: composition run backwards): a real-linear map with $(ab)^* = b^*a^*$ and $(a^*)^* = a$ for all a, b." (Lemma 5.1)

norm form N

"Then every $a \in \mathcal{A}$ is $x + u$ with x real and $u^* = -u$; for every such u the square u^2 is real; and $N(a) = aa^* = a^*a = x^2 - u^2$ is real." (Lemma 5.1)

unit group (of $\mathbb{R} \oplus \mathbb{R}u$**)**

"For $u \neq 0$ with $u^* = -u$ the elements 1 and u are linearly independent, and the unit group of $\mathbb{R} \oplus \mathbb{R}u$ is the curve $\{x + yu : x^2 - u^2y^2 = 1\}$ in the plane with coordinates (x, y)." (Lemma 5.1)

Q_8**;** $\mathbb{C}_i, \mathbb{C}_j, \mathbb{C}_k$**;** V**;** Ad_u

"The subalgebras of $\mathbb{H}$ generated by a single unit of $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ are $\mathbb{R}$, $\mathbb{C}_i$, $\mathbb{C}_j$, $\mathbb{C}_k$, in bijection with $Q_8/\{\pm 1\} \cong \mathbb{Z}_2 \times \mathbb{Z}_2$: there are exactly four. The group of inner automorphisms by units, $V = \{\mathrm{id}, \mathrm{Ad}_i, \mathrm{Ad}_j, \mathrm{Ad}_k\}$ with $\mathrm{Ad}_u(q) = uqu^{-1}$, is the same Klein four-group." (Proposition 5.3)

w**;** Ad_w**; the algebra at a point**

"let the algebra at a point be the subalgebra of $\mathbb{H}$ fixed by those of the fold and one step of the drive that leave the point in place, the local form of ι being conjugation at the drive-fixed corner and the inner automorphism by the surviving unit at the three cycled corners, and the drive acting as the order-three inner automorphism Ad_w, $w = \frac{1}{2}(1 + i + j + k)$" (Proposition 5.6) The surviving unit at a cycled corner is the unit of its algebra: i at P_2, j at P_3, k at P_4.

pillowcase

"The quotient T^2/ι is the pillowcase orbifold: a sphere with four cone points of cone angle π, the unique closed Euclidean two-orbifold of signature $(2, 2, 2, 2)$." (Theorem 5.7)

covering torus E

"Let E be the covering torus of the pillowcase with a complex structure for which the fold is $\iota(z) = -z$, and let $P_1, \ldots, P_4$ be the four corners." (Section 7.4)

twist of the fibre; flux

"A *twist of the fibre* of *flux* $d \geq 1$ is the line $\mathcal{O}(D)$ on E of a divisor $D = \sum a_i P_i$ on the corners, with $a_i \geq 0$ and $\sum a_i = d$." (Definition 7.4)

half-turn twist

"A twist of flux zero is a flat line with holonomy ± 1: the trivial one, or one of the three *half-turn twists*." (Definition 7.4)

state at rest

"A *state at rest* is a holomorphic section of the twist: for $d \geq 1$, a function f on the covering torus E whose poles are bounded by D." (Definition 7.4)

first lift; second lift (of the fold)

"The fold acts on states at rest by $f(z) \mapsto f(-z)$ or by $f(z) \mapsto -f(-z)$, its two lifts, called here the first and the second." (Definition 7.4)

state on the pillowcase

"A *state on the pillowcase*, for a given lift, is a state at rest that the lift leaves unchanged." (Definition 7.4)

corner sign

"The *corner sign* of the twist at P_i is $(-1)^{a_i}$, the action of the first lift on the fibre of $\mathcal{O}(D)$ there." (Definition 7.4)

odd corner

"A corner is *odd* if its sign differs from that of the other three." (Definition 7.4)

structure (of a twist)

"The *structure* of a twist is its flux together with its corner signs up to the overall sign or, for flux zero, its holonomy." (Definition 7.4)

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